Explicit Stability of the Weak Simplex Bound

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Abstract

Equal-energy Gaussian signaling uses the regular simplex as an optimal reference when the signal dimension permits it. Quantifying geometric recovery from a small decoding deficit requires a bound that remains valid near singular Gram matrices. We give explicit global quadratic bounds for Gram and vertex distance at every fixed positive channel parameter, together with a separate centroid penalty. A configuration-metric expansion separates two in-plane sectors from a transverse centroid sector and identifies the local least-loss directions. We prove unique ordered channel-parameter crossings and identify the exact minimizers on all sufficiently small Procrustes shells in the strict-normal regime, uniformly in the ambient dimension. Their minimal deficit has an explicit quartic expansion with a sixth-order remainder. The global bounds, mean-width stability, transition and small-shell results, including the remainder, are formalized in Lean. For four- and 16-signal constellations, directional coefficients predict central perturbation losses more accurately in aggregate than two scalar controls across 108 tested cells. A separate 20-run learning comparison attributes improved matched-decoder performance to ordinary simplex Gram regularization, rather than to a new optimization method. The evaluated global recovery constants range from approximately 10^{-39} to 10^{-234} and give no practical finite-precision geometry certificate. The results provide an explicit mathematical rate and a directional local diagnostic, while practical certification requires sharper constants and effective numerical bounds for the local radius and remainder constant.

1 Setting and the quantitative question

Fix $m \geq 2$, put $n = m - 1$, $\alpha = n/m$, $J = \mathbf{1}\mathbf{1}^{\mathsf{T}}$ and $P = I_m - J/m$. The regular unit simplex has Gram matrix $G_\Delta = P/\alpha$. For a correlation matrix G (positive semidefinite with unit diagonal), define

$$R = \alpha G + J/m, \qquad \mathcal{A}_m = \{R : \operatorname{diag} R = \mathbf{1},\ R - J/m \succeq 0\}. \tag{1}$$

Thus $G = G_\Delta$ corresponds to $R = I_m$. Write

$$F_R(c) = \mathbb{P}\{X_i \leq c \text{ for all } i\}, \qquad X \sim N(0, R), \quad D_R(c) = F_R(c) - \Phi(c)^m,$$

where Φ and ϕ are the standard normal distribution and density. We use the comparison $D_R(c) \geq 0$ proved in [1], as well as its centered product inequality and variational construction. Pointwise strictness was established independently by Su et al. [2] and by Mulgund, whose v4 explicitly records this parallel development [7, Remark 2.3 and Appendix B]. Su et al. also explicitly suggest a quantitative deficit estimate. The present argument replaces a strict positive gap by a computable lower bound. We cite v2 for the particular product and variational statements used in the proof, not as a description of the current state of Mulgund's work.

For unit signals $x_1, \ldots, x_m$ in $\mathbb{R}^d$, let $G_{ij} = \langle x_i, x_j \rangle$ and consider $Y = \lambda x_I + Z$, with I uniform, $Z \sim N(0, I_d)$, and $\lambda > 0$. Maximum-likelihood decoding gives the identity

$$P_{\mathrm{c}}(G; \lambda) = \frac{e^{-\lambda^2/2}}{m} \mathbb{E} e^{\lambda \max_i \xi_i}, \qquad \xi \sim N(0, G). \tag{2}$$

All deficits below compare to the unit centered regular simplex at the same operating parameter. The integer m is the number of signals, and d is the ambient dimension of the signal space. Decoding constants and unnormalized expected-maximum constants depend on m and, for decoding, on λ, but not on d. Only those constants are called uniform in the signal-space dimension. The normalized mean-width lower bound (22) retains the factor $2/\mathbb{E}\,\|g\|$ for $g \sim N(0, I_d)$, and that prefactor depends on d. No positive lower constant uniform in m, and no positive lower constant uniform over all positive channel parameters, is asserted.

Local optimality of simplex codes is classical. Litvak attributes fixed-parameter local maximality to Balakrishnan [4, Theorem 2.5], whose original article is [8]. Weber already expresses the general second variation through repeated, adjacent and disjoint edge pairs [12, Lemma 5.8, equation (5.102)]. His three quadratic invariants are the squared edge norm, squared total row sum and sum of squared row sums. On his feasible zero-total-sum face, the row sums vanish and (5.105) gives the restricted form used in Theorem 5.5. Equations (5.107)–(5.108) integrate the distribution-function variation into the decoding functional. Thus edge-type decomposition and local simplex optimality are established antecedents of the calculation below. We claim no priority for these local ideas or for the existence of a Hessian. Our global quantitative contribution is the specified rate and its conversion into configuration distance. The local formulas supply explicit normalization and feasible-sequence criteria for using that rate. Section 7 additionally proves unique channel transitions and an exact, rather than only second-order, optimizer on small shells in the strict-normal regime. Its complete analytic and geometric proofs are in Appendices A and B; the precise scope of the Lean verification is given in Appendix C.

Mean-width stability under John/Löwner ellipsoid constraints [5] concerns a different admissible class. The unsmoothed comparison in [6, Theorem 1.1] dominates Gaussian maxima at every real threshold and characterizes equality at positive thresholds. Its equality statement does not extend to every nonpositive threshold. We use the cited comparison and centered product inequality as inputs. Quantitative Gaussian correlation inequalities also predate this work [10]. Our rectangle estimate is a specialized input.

Friedland, Guédon, and Souli [11] prove an explicit stability theorem for a different functional. For $\beta \in \mathbb{R}$ they set $Z_\beta(X) = \sum_{i=1}^N e^{\beta X_i}$, where X is centered Gaussian with unit variances and correlation matrix $C \in E_N$. The set E_N consists of all $N \times N$ correlation matrices, and Δ_N is the regular-simplex correlation. That matrix equals G_Δ. The symbol β in this discussion is theirs, not the scalar in (5). Their coefficients $\mu_{N,q,\beta}$ and $\eta_{N,q,\beta}$ are not the constants μ and η in (15) and (13).

Their Theorem 1.1 is a phase transition of generalized L_q-means, for every real q, every real β, and every $C \in E_N$. The simplex correlation maximizes $\|Z_\beta(X)\|_q$ when $q < 1$. The mean equals $Ne^{\beta^2/2}$ for every such C when $q = 1$. The simplex correlation minimizes the mean when $q > 1$. The quantitative statement is narrower. Theorem 1.3 requires $q > 1$ and $\beta \neq 0$, and concludes

$$\mathbb{E}[Z_\beta(X)^q] - \mathbb{E}[Z_\beta(S)^q] \geq \eta_{N,q,\beta}\, \mathbf{1}^{\mathsf{T}} C \mathbf{1} + \frac{\mu_{N,q,\beta}}{2} \|C - \Delta_N\|_F^2 ,$$

where $S \sim N(0, \Delta_N)$ and both coefficients are positive. The summand $\eta_{N,q,\beta}\, \mathbf{1}^{\mathsf{T}} C \mathbf{1}$ measures centroid defect, and the Frobenius summand measures shape defect. The same modulus $\mu_{N,q,\beta}$, their (15), is the strong-convexity modulus of $C \mapsto \mathbb{E}[Z_\beta(X)^q]$ on all of E_N in their Theorem 1.2. Their (15) and (16) vanish at $\beta = 0$, so both the modulus and the two stability terms are zero in that case. Theorem 1.1 remains valid at $\beta = 0$, where Z_0 is the constant N.

The estimates below are not those statements. Since $\alpha G_\Delta = P = I_m - J/m$, the normalization (1) gives $R - I_m = \alpha(G - G_\Delta)$ and identifies $\mathcal{A}_m$ with the image of E_m. Since $\Delta_m = G_\Delta$, it follows that $\|R - I_m\|_F = \alpha \|G - \Delta_m\|_F$. The constant $\kappa_{m,[u,v]}$ depends on a compact threshold interval and is not $\mu_{N,q,\beta}$. Theorem 1.3 controls moments of order

$q > 1$ of Z_β on E_N. The deficit $D_R(c)$ is an orthant probability on $\mathcal{A}_m$, uniform for c in a compact interval, and the bound includes singular R. The inequality $D_R(c) \geq 0$ has the direction of their Corollary 1.5 applied to the indicator of $(-\infty, c]$. That corollary assumes a correlation matrix R with $R - J/N \succeq 0$, which is the class $\mathcal{A}_N$ with their N equal to the number of signals m. Remark 1.6 records the resulting non-strict bound as generalizing an orthant inequality of Mulgund. The corollary does not give a quadratic rate. For a Gram matrix of unit vectors, their centroid summand and Lemma 4 both control $\|\sum_i x_i\|^2 = \mathbf{1}^\mathsf{T} G \mathbf{1}$, through different functionals and different coefficients. Decoding and mean-width stability below concern $\mathbb{E}e^{\lambda \max_i \xi_i}$ and $\mathbb{E}\max_i \xi_i$. They record that $\beta^{-1}\mathbb{E}\log Z_\beta$ tends to $\mathbb{E}\max_i X_i$ as $\beta \to +\infty$. At $q = 0$, Theorem 1.1 compares $\mathbb{E}\log Z_\beta$ at fixed β. Theorem 1.3 covers neither statement, because it assumes $q > 1$. The full-class local liminf, including the split between $m = 3$ and $m \geq 4$, is not a conclusion of Theorem 1.1 or Theorem 1.3.

Proposition 1 (What compactness supplies)**.** *Let $\mathcal{K}$ be a compact metric space, $Z \subset \mathcal{K}$ a nonempty closed set, and $f : \mathcal{K} \to [0, \infty)$ continuous with zero set exactly Z. Then $f(x_j) \to 0$ implies $d(x_j, Z) \to 0$. If, in addition, $f(x) \geq a\, d(x, Z)^2$ whenever $d(x, Z) < \rho$, with $a, \rho > 0$, then there exists $b > 0$ such that $f(x) \geq b\, d(x, Z)^2$ on $\mathcal{K}$.*

Proof. For each $\epsilon > 0$, compactness gives a positive minimum of f on $\{d(x, Z) \geq \epsilon\}$ whenever that set is nonempty. This proves the first assertion. For the second, let $M = \max_{\mathcal{K}} d(x, Z)$. If $M = 0$ or the set $\{d(x, Z) \geq \rho\}$ is empty, the assertion is immediate. Otherwise its minimum η is positive, and $b = \min\{a, \eta/M^2\}$ works. □

For fixed m, the correlation-matrix domain is compact. The analogous vertex formulation uses a fixed ambient space and the closed simplex orbit. Thus a nonexplicit global quadratic bound is a standard consequence of uniqueness and a local quadratic bound in the *same* metric. Uniqueness and compactness alone give no power law. Nor are the Gram and vertex metrics locally equivalent at the rank-deficient simplex, as Section 6 shows through an explicit transverse example. These distinctions delimit the role of our global proof and constants.

2 Explicit constants and the global comparison

Define the scalar functions

$$r(s) = \frac{\phi(s)}{\Phi(s)}, \qquad H(s) = s + r(s), \qquad \Psi(H(s)) = \log \Phi(s) + \frac{1}{2} r(s)^2. \tag{3}$$

The map $H : \mathbb{R} \to (0, \infty)$ is an increasing bijection. Moreover $\Psi : (0, \infty) \to (-\infty, 0)$ is an increasing, strictly concave bijection, with

$$\Psi'(H(s)) = r(s), \quad \Psi''(H(s)) = -\frac{r(s)H(s)}{1 - r(s)H(s)} < 0. \tag{4}$$

The denominator is the variance of a standard normal conditioned on $Z \leq s$. These scalar identities also fix the meaning of the one-dimensional inverses and maximizations used in our constants.

Fix a finite interval $[u, v]$, allowing $u = v$, and set

$$\begin{aligned} &L_0 = m \log \Phi(u), && \beta = \Psi^{-1}(L_0), && A_0 = \Psi'(\beta), \\ &B_0 = v + \sqrt{-2L_0/m} + \sqrt{-2\alpha L_0}, && \ell = \beta/\sqrt{2}, && T = \sqrt{2}B_0 + \ell. \end{aligned} \tag{5}$$

All these quantities are finite, and β, ℓ, B_0, T, A_0 are positive. For $t > 0$ put $b(t) = 2\Phi(t) - 1$. Define

$$K_{m,[u,v]} = \frac{e^{-mA_0\beta}\ell^2 e^{-4T^2/3} b(\ell)^{2m-2}}{4\pi m(m-1)}, \qquad \kappa_{m,[u,v]} = \frac{\Phi(u)^m K_{m,[u,v]}}{1 + \sqrt{1 + K_{m,[u,v]} m(m-1)}}. \tag{6}$$

Theorem 2 (Global quantitative comparison)**.** *For every $R \in \mathcal{A}_m$ and $c \in [u, v]$,*

$$D_R(c) \geq \Phi(u)^m \left(\sqrt{1 + K_{m,[u,v]} \, \|R - I_m\|_F^2} - 1 \right) \geq \kappa_{m,[u,v]} \, \|R - I_m\|_F^2 \, . \tag{7}$$

The conclusion includes singular R.

We first quantify the two-coordinate input. Its conversion to a full rectangle uses the Gaussian correlation inequality [3].

Lemma 3 (Quantitative rectangle gap)**.** *Let $V \sim N(0, S)$, where S is any $m \times m$ correlation matrix, possibly singular. If $t_i \in [\ell, T]$, with $0 < \ell \leq T < \infty$, then*

$$\mathbb{P}\{|V_i| \leq t_i \ \forall i\} - \prod_i b(t_i) \geq \frac{\ell^2 e^{-4T^2/3} b(\ell)^{m-2}}{4\pi m(m-1)} \, \|S - I_m\|_F^2 \, . \tag{8}$$

Proof. For a bivariate standard normal with correlation ρ, let $B_\rho(a, b) = \mathbb{P}\{|V_1| \leq a, |V_2| \leq b\}$. This is even in ρ. Differentiating the Gaussian integral (equivalently, integrating $\partial_\rho f_\rho = \partial_x \partial_y f_\rho$) gives, for $0 < \rho < 1$,

$$\partial_\rho B_\rho(a, b) = \frac{2}{\pi\sqrt{1-\rho^2}} e^{-(a^2+b^2)/(2(1-\rho^2))} \sinh\left(\frac{\rho a b}{1-\rho^2} \right).$$

For $a, b \in [\ell, T]$ and $0 \leq \rho \leq 1/2$, this derivative is at least $(2\ell^2/\pi) e^{-4T^2/3} \rho$. It is nonnegative on $[0, 1)$, so integration gives

$$B_\rho(a, b) - b(a)b(b) \geq \frac{\ell^2}{\pi} e^{-4T^2/3} \min\{|\rho|, 1/2\}^2 \geq \frac{\ell^2}{4\pi} e^{-4T^2/3} \rho^2.$$

Continuity extends this non-strict quantitative inequality to $|\rho| = 1$; the rectangle boundary has zero probability since the marginals have no atoms. Choose $i < j$ with maximal S_{ij}^2. Group this two-coordinate rectangle with the remaining symmetric slabs using the Gaussian correlation inequality. Its deficit is multiplied by at least $b(\ell)^{m-2}$. Finally $\max_{i<j} S_{ij}^2 \geq \|S - I_m\|_F^2 / [m(m-1)]$. For singular S, apply the correlation inequality to the linear preimages of these sets under a representation $V = TW$ with nondegenerate W. □

Proof of Theorem 2. Represent $G = (R - J/m)/\alpha$ by unit vectors x_i. We recall precisely the two inputs from [1]. The centered product inequality says

$$\mathbb{E} \prod_i f_i(Y_i) \geq \prod_i \int f_i \, d\gamma_1 \tag{9}$$

for any Gaussian correlation matrix of Y, and bounded nonnegative log-concave factors with positive Gaussian integrals and zero Gaussian first moments. The adaptive tilts arise from maximizing

$$\mathcal{J}_c(q, w) = \sum_i \Psi(q + \alpha \langle x_i, w \rangle) - \frac{\alpha}{2} \|w\|^2 - \frac{m}{2}(q - c)^2$$

on the domain where all scalar arguments are positive. Its maximizer has

$$b_i = H(s_i) = c + (Ra)_i, \quad a_i = r(s_i), \qquad \mathcal{J}_* \geq m \log \Phi(c) \geq L_0. \tag{10}$$

For clarity, stationarity of the strictly concave variational functional gives $w_* = \sum_i a_i x_i$ and $q_* - c = m^{-1} \sum_i a_i$, and hence

$$\mathcal{J}_* = \sum_i \Psi(b_i) - \tfrac{1}{2} a^\mathsf{T} R a.$$

The maximizer exists in the positive-argument domain: the quadratic terms control infinity, whereas $\Psi(y) \to -\infty$ as $y \downarrow 0$. We now extract bounds not requiring any inverse of R. Since every summand $\Psi(b_i)$ and both negative quadratic terms are nonpositive, $\Psi(b_i) \geq L_0$. Thus $b_i \geq \beta$ and $a_i \leq A_0$. The quadratic penalty is at most $-L_0$, whence

$$|q - c| \leq \sqrt{-2L_0/m}, \quad \|w\| \leq \sqrt{-2L_0/\alpha}, \quad b_i \leq B_0.$$

In particular $B_0 > 0$.

Let $f_i(z) = e^{a_i z}\mathbf{1}_{z \leq b_i}$, $\mu_i = \int f_i \, d\gamma_1 = e^{\Psi(b_i)} < 1$, and $h_i = f_i/\mu_i$. They are centered in Gaussian first moment. For

$$(Dh)(u) = \int h((u+v)/\sqrt{2})h((u-v)/\sqrt{2}) \, d\gamma_1(v),$$

write $Q_0 = \mathbb{E}_R \prod_i h_i(X_i)$ and $Q_1 = \mathbb{E}_R \prod_i (Dh_i)(X_i)$. Taking independent $X, X' \sim N(0, R)$ and setting $U = (X + X')/\sqrt{2}$, $V = (X - X')/\sqrt{2}$, the vectors U, V are independent with covariance R. Conditional on $U = u$, the factors become

$$h_i((u_i + v_i)/\sqrt{2})h_i((u_i - v_i)/\sqrt{2}) = \mu_i^{-2} e^{\sqrt{2}a_i u_i} \mathbf{1}_{|v_i| \leq \sqrt{2}b_i - u_i}. \tag{11}$$

On $\mathcal{B} = \{|U_i| \leq \ell \ \forall i\}$ all radii belong to $[\ell, T]$. The product of the weights in (11) is at least $e^{-\sqrt{2}mA_0\ell} = e^{-mA_0\beta}$. The symmetric rectangle inequality gives $\mathbb{P}(\mathcal{B}) \geq b(\ell)^m$, even for singular R. Lemma 3 therefore yields

$$Q_0^2 - Q_1 \geq K_{m,[u,v]} \, \|R - I_m\|_F^2 . \tag{12}$$

Outside $\mathcal{B}$ the conditional difference remains nonnegative: use the ordinary rectangle inequality when all radii are positive; if a radius is negative both products vanish, and a zero radius also has zero mass because every coordinate marginal is continuous.

Each Dh_i is bounded and log-concave, has unit Gaussian integral and zero Gaussian first moment. The integral and moment assertions follow by the same orthogonal change of two independent scalar Gaussian variables; log-concavity follows by integration of a log-concave function. Thus (9) gives $Q_1 \geq 1$. The Gaussian change of measure and (10) give the exact identity

$$F_R(c) = e^{\mathcal{J}_*} Q_0.$$

This identity is valid for singular R as well: write $X = TZ$ with Z standard Gaussian in the rank of R and tilt by $T^{\mathsf{T}}a$. The shifted vector is $X + Ra$, and $b - Ra = c\mathbf{1}$; no inverse covariance or full-dimensional density is used. Consequently $F_R(c) \geq \Phi(c)^m \sqrt{1 + K_{m,[u,v]} \, \|R - I_m\|_F^2}$. Subtract $\Phi(c)^m$ and use $\Phi(c)^m \geq \Phi(u)^m$. Finally $\|R - I_m\|_F^2 \leq m(m-1)$, since $|R_{ij}| \leq 1$. Rationalizing the square root proves (7). □

3 A centroid estimate and decoding stability

For the rest of the global consequences use the interval $[0, 1]$ and abbreviate $\kappa = \kappa_{m,[0,1]}$. Define

$$M_0 = \max_{y \in [H(0)/2, \, 3H(1)/2]} (-\Psi''(y)), \quad \tau = \min\left\{ \frac{H(0)}{2\alpha m}, \frac{r(1)}{1 + M_0 \alpha m} \right\}, \quad \eta = 2^{-m} \frac{\alpha r(1) \tau}{2} > 0. \tag{13}$$

Lemma 4 (Centroid penalty)**.** *For unit vectors x_i with Gram matrix G and R as in* (1)*, set $s_X = \sum_i x_i$. For $0 \leq c \leq 1$,*

$$D_R(c) \geq \eta \, \|s_X\|^2 . \tag{14}$$

Proof. In the variational functional use $q = H(c)$ and $w = \tau s_X$. Since $\|s_X\| \le m$, the scalar displacements $z_i = \alpha\tau\langle x_i, s_X\rangle$ obey $|z_i| \le H(0)/2$. Taylor's theorem on the interval in (13) gives

$$\Psi(H(c) + z_i) \ge \Psi(H(c)) + r(c)z_i - M_0 z_i^2/2.$$

Now $\sum_i z_i = \alpha\tau \|s_X\|^2$ and $\sum_i z_i^2 \le \alpha^2\tau^2 m \|s_X\|^2$. Subtracting the value at $(H(c), 0)$ gives

$$\mathcal{J}_* - m \log \Phi(c) \ge \alpha \left(r(c)\tau - \frac{\tau^2}{2}(1 + M_0\alpha m) \right) \|s_X\|^2 \ge \frac{\alpha r(1)\tau}{2} \|s_X\|^2 .$$

Use $F_R(c) \ge e^{\mathcal{J}_*}$, $e^z - 1 \ge z$, and $\Phi(c)^m \ge 2^{-m}$. □

Let $\mu = \lambda/\sqrt{\alpha}$ and put

$$W_{m,\lambda} = \frac{e^{-\mu^2/2}}{m}(e^\mu - 1), \qquad c_g = W_{m,\lambda}\kappa\alpha^2, \quad c_s = W_{m,\lambda}\eta, \quad c_P = \left(\frac{\alpha}{c_g} + \frac{1}{mc_s} \right)^{-1}. \tag{15}$$

These constants are positive for every fixed $m \ge 2$ and $\lambda > 0$.

Theorem 5 (Quantitative recovery of a simplex)**.** *Let G be the Gram matrix of m unit signals and $\delta_\lambda(G) = P_c(G_\Delta; \lambda) - P_c(G; \lambda)$. Then*

$$\delta_\lambda(G) \ge c_g \|G - G_\Delta\|_F^2 , \qquad \delta_\lambda(G) \ge c_s \left\|\textstyle\sum_i x_i\right\|^2 . \tag{16}$$

If the ambient dimension is at least $m - 1$, fix a regular simplex matrix $V = [v_1 \ \cdots \ v_m]$ in that space, and define

$$d_P(X, \Delta)^2 = \min_{U \in O(d)} \sum_i \|x_i - Uv_i\|^2 .$$

Then

$$\delta_\lambda(G) \ge c_P d_P(X, \Delta)^2. \tag{17}$$

In particular $\delta_\lambda(G) \le \varepsilon$ implies $d_P(X, \Delta) \le \sqrt{\varepsilon/c_P}$.

Proof. Add an independent scalar $B \sim N(0, 1/n)$ and set $X_i' = \sqrt{\alpha}(\xi_i + B)$. Its covariance is R and

$$\mathbb{E}e^{\mu \max_i X_i'} = e^{\lambda^2/(2n)} \mathbb{E}e^{\lambda \max_i \xi_i}.$$

For the simplex the transformed coordinates are independent standard normals. All the exponential moments are finite. Integrating their tail probabilities gives

$$\mathbb{E}e^{\mu \max_i Z_i} - \mathbb{E}e^{\mu \max_i X_i'} = \mu \int_{\mathbb{R}} e^{\mu c} D_R(c)\, dc.$$

The integrand is nonnegative everywhere by the known comparison. Restrict it to $[0, 1]$, use Theorem 2 or Lemma 4, and then use (2). The prefactor becomes $m^{-1}e^{-\lambda^2/2 - \lambda^2/(2n)} = m^{-1}e^{-\mu^2/2}$. This proves (16).

For completeness, the geometric conversion is quantitative linear algebra. Let $X = [x_1 \ \cdots \ x_m]$ and let $\lambda_1, \ldots, \lambda_n$ be the eigenvalues of PGP on $\mathbf{1}^\perp$, including zeros. The cross matrix XV^T has nonzero singular values $\sqrt{\lambda_j/\alpha}$, with zeros added as necessary. The orthogonal Procrustes formula follows by maximizing $\operatorname{tr}(U^\mathsf{T}XV^\mathsf{T})$ over $O(d)$. Alignment and the orthogonality of centered and constant columns give

$$\begin{aligned} d_P(X, \Delta)^2 &= \frac{\|s_X\|^2}{m} + \sum_{j=1}^{n} (\sqrt{\lambda_j} - \alpha^{-1/2})^2 \\ &\le \frac{\|s_X\|^2}{m} + \alpha \|PGP - G_\Delta\|_F^2 \le \frac{\|s_X\|^2}{m} + \alpha \|G - G_\Delta\|_F^2 . \end{aligned}$$

The first inequality uses $(\sqrt{t}-\sqrt{a})^2 \le (t-a)^2/a$ for $t \ge 0, a > 0$. Applying the two separate deficit bounds proves (17). Relabeling is already included: every permutation of regular-simplex vertices extends to an orthogonal symmetry. □

Corollary 6 (Insufficient dimension and unused energy)**.** *If the unit-signal Gram matrix has rank at most $d < m-1$, with $d \ge 1$, then*

$$\delta_\lambda(G) \ge c_g m^2 \left(\frac{1}{d} - \frac{1}{m-1}\right). \tag{18}$$

For arbitrary signals $\|x_i\| \le 1$, define the deficit using the actual maximum-likelihood success probability (not the equal-energy formula (2)). Then, regardless of whether the original channel permits an additional use,

$$P_{\mathrm{c}}(G_\Delta;\lambda) - P_{\mathrm{c}}(\{x_i\};\lambda) \ge c_s \sum_i (1 - \|x_i\|^2). \tag{19}$$

Proof. Using $\operatorname{tr} G = m$,

$$\|G - G_\Delta\|_F^2 = \operatorname{tr}(G^2) - \frac{m^2}{m-1} + \frac{2}{m-1}\mathbf{1}^{\mathsf{T}} G \mathbf{1} \ge \frac{m^2}{d} - \frac{m^2}{m-1}.$$

Here $\operatorname{tr}(G^2) \ge (\operatorname{tr} G)^2/d$ and $\mathbf{1}^{\mathsf{T}} G \mathbf{1} \ge 0$. For the energy claim, use the following lift only as a mathematical comparison (it imposes no additional-channel-use assumption on the code): lift the codewords to $\widetilde{x}_i = (x_i, q_i)$, $q_i = \sqrt{1 - \|x_i\|^2} \ge 0$. The augmented signals have unit norm, and their optimal success probability is no smaller because the receiver may ignore the additional observation. Apply the centroid inequality in (16) and note $\|\sum_i \widetilde{x}_i\|^2 \ge (\sum_i q_i)^2 \ge \sum_i q_i^2$. □

4 Mean width of arbitrary inscribed simplices

Set

$$b_g = \kappa \alpha^{3/2}, \qquad b_s = \eta/\sqrt{\alpha}. \tag{20}$$

For any correlation matrix G the same distribution-function integration, without the exponential weight, proves

$$\mathbb{E}\max_i \xi_i^\Delta - \mathbb{E}\max_i \xi_i \ge b_g \|G - G_\Delta\|_F^2, \qquad \mathbb{E}\max_i \xi_i^\Delta - \mathbb{E}\max_i \xi_i \ge b_s \left\|\textstyle\sum_i x_i\right\|^2. \tag{21}$$

Indeed the expectation gap of the transformed maxima is $\int_{\mathbb{R}} D_R(c)\,dc$, and the common added Gaussian has mean zero.

Theorem 7 (Mean-width stability in a ball)**.** *Let $y_1, \dots, y_m \in B_2^d$ with $d \ge m-1$, let $K = \operatorname{conv}\{y_i\}$, and let Δ be a unit regular simplex in the same ambient space. Use normalized spherical mean width, so that $w(K) = 2\mathbb{E}\max_i \langle g, y_i\rangle / \mathbb{E}\|g\|$ for $g \sim N(0, I_d)$. Then*

$$w(\Delta) - w(K) \ge \frac{2}{\mathbb{E}\|g\|} \left(\frac{\alpha}{b_g} + \frac{1}{b_s}\right)^{-1} d_P([y_i], \Delta)^2. \tag{22}$$

Also the unnormalized expected-maximum deficit is at least $b_s \sum_i (1 - \|y_i\|^2)$.

Proof. Put $d_i = 1 - \|y_i\|^2$ and lift to unit vectors $\widehat{y}_i = (y_i, \sqrt{d_i} e_i)$ using mutually orthogonal new coordinates. Conditional Jensen implies that adding these independent mean-zero noises cannot decrease the expected maximum. Write $G = (\langle y_i, y_j\rangle)$, $\widehat{G} = G + \operatorname{diag}(d_i)$, $T_g = \left\|\widehat{G} - G_\Delta\right\|_F^2$, and $T_s = \left\|\sum_i y_i\right\|^2 + \sum_i d_i$. The original expected-maximum deficit δ_w is at least $b_g T_g$ and at

least $b_s T_s$ by (21). The Procrustes argument in the previous proof applies to vectors of any lengths, and hence

$$\begin{aligned} d_P([y_i], \Delta)^2 &\le \alpha \|G - G_\Delta\|_F^2 + \left\|\sum_i y_i\right\|^2 / m \\ &= \alpha \left(T_g + \sum_i d_i^2 \right) + \left\|\sum_i y_i\right\|^2 / m \le \alpha T_g + T_s, \end{aligned}$$

since $0 \le d_i \le 1$. Thus $d_P^2 \le (\alpha/b_g + 1/b_s)\delta_w$. The polar decomposition of a standard Gaussian gives (22). □

5 Sharp local coefficients and optimal exponents

The explicit global constants are deliberately conservative. The exact local coefficient is much simpler and identifies a distinction between $m = 3$ and $m \ge 4$.

Theorem 8 (Sharp local common-threshold coefficient)**.** *Fix a finite c, and write $p = \Phi(c)$, $q = r(c)$, $h = c + q$, and $k = \phi(c)^2 p^{m-2}$. For $m \ge 3$,*

$$\liminf_{\substack{R \to I_m,\ R \in \mathcal{A}_m \\ R \ne I_m}} \frac{D_R(c)}{\|R - I_m\|_F^2} = \begin{cases} k(1 + ch)/4, & m = 3, \\ kh^2/4, & m \ge 4. \end{cases} \tag{23}$$

Both constants are positive. For $m = 2$ the sharp behavior is linear: $D_R(c)/\|R - I_2\|_F \to \phi(c)^2/\sqrt{2}$.

Proof. Put $A = R - I_m$, $S = \sum_{i<j} A_{ij}$, $Q = \sum_{i<j} A_{ij}^2 = \|A\|_F^2/2$, and $r_A = A\mathbf{1}$. The Gaussian density is smooth in A near zero and its first three derivatives are bounded by an integrable polynomial times a Gaussian. Differentiation under the integral over the fixed lower orthant is therefore justified. For off-diagonal covariance entries, the first derivative at independence is k. The second-order Taylor contributions for a repeated edge, two edges sharing one endpoint, and two disjoint edges are respectively

$$\tfrac{1}{2} c^2 \phi(c)^2 p^{m-2} A_{ij}^2, \quad -c\phi(c)^3 p^{m-3} A_{ij} A_{ik}, \quad \phi(c)^4 p^{m-4} A_{ij} A_{k\ell}.$$

Each unordered pair of distinct edges is counted once. The sums of products over adjacent and disjoint pairs are $\|r_A\|^2/2 - Q$ and $(S^2 + Q - \|r_A\|^2)/2$, respectively. Consequently

$$\frac{D_{I_m + A}(c)}{k} = S + \frac{h^2}{2} Q - \frac{qh}{2} \|r_A\|^2 + \frac{q^2}{2} S^2 + O(\|A\|_F^3). \tag{24}$$

For $m = 3$ there are no disjoint pairs; the displayed combined formula still holds because the corresponding combinatorial sum is zero.

Feasibility means $L = P + A \succeq 0$. Since $L\mathbf{1} = r_A$, the spectral inequality $L^2 \preceq \lambda_{\max}(L) L$ gives

$$S \ge 0, \qquad \|r_A\|^2 \le 2(1 + \|A\|_{\mathrm{op}}) S. \tag{25}$$

Also $\sigma = 1 - qh > 0$ is a truncated-normal variance. Equations (24)–(25) imply, for sufficiently small feasible A,

$$D_{I_m + A}(c) \ge k \left(\frac{\sigma}{2} S + \frac{h^2}{4} Q \right).$$

Thus sequences with $S/Q \to \infty$ have an infinite deficit quotient. On sequences with $S = O(Q)$, (25) yields $S \ge \|r_A\|^2/2 + o(Q)$ and hence

$$D_{I_m + A}(c)/k \ge \tfrac{1}{2} h^2 Q + \tfrac{1}{2} \sigma \|r_A\|^2 + o(Q). \tag{26}$$

For $m \geq 4$ this gives the lower coefficient $kh^2/4$. It is attained by $A = tD$, where D is symmetric with $D_{12} = D_{34} = 1$, $D_{23} = D_{14} = -1$, and all other entries zero. Then $D\mathbf{1} = 0$, so $P + tD \succeq 0$ for sufficiently small $|t|$.

For $m = 3$ there is the identity $\|r_A\|^2 = Q + S^2$. Equation (26) gives the lower coefficient $k(h^2 + \sigma)/4 = k(1 + ch)/4$. To attain it, take a regular simplex $v_i \in \mathbb{R}^{m-1}$, a unit vector $u \perp v_1$, and replace only v_1 by $v_1(t) = \cos(t)v_1 + \sin(t)u$. For the resulting $A = \alpha(G(t) - G_\Delta)$ one computes exactly

$$S = \alpha(1 - \cos t), \quad Q = \frac{\alpha^2}{m-1}((1 - \cos t)^2 + m \sin^2 t), \quad \|r_A\|^2 = 2S.$$

Thus $Q = \alpha t^2 + O(t^4)$ and the desired quotient follows from (24). Positivity follows from $h^2 + \sigma > 0$. For $m = 2$, feasibility is $R_{12} = \rho \geq 0$ and the first derivative at $\rho = 0$ is $\phi(c)^2$, whereas $\|R - I_2\|_F = \sqrt{2}\rho$. □

Corollary 9 (Optimal powers)**.** *For $m \geq 3$, the power two of the Frobenius distance in the common-threshold bound and in decoding Gram stability cannot be replaced by any smaller positive power with a uniform positive constant near the simplex. For $m \geq 3$, the power two of the coordinate distance in* (17) *is likewise optimal, including when signals are restricted to dimension $m - 1$.*

Proof. Use the one-vertex rotation from the proof of Theorem 8. It gives $\|G(t) - G_\Delta\|_F = \Theta(|t|)$ and $S = O(t^2)$. The normalized covariance $R(t)$ remains near I_m, where expectations of $e^{\mu \max_i X_i}$ are twice differentiable by Gaussian-density domination. Permutation symmetry makes their first derivative a constant times the sum of the off-diagonal perturbations. That sum has zero first derivative along this path. The decoding deficit is therefore $O(t^2)$, and the lower bound in Theorem 5 makes it $\Theta(t^2)$. The same argument applies to the fixed-threshold deficit. Moreover $\|G - G_\Delta\|_F \leq 2\sqrt{m}\, d_P(X, \Delta)$ for unit columns, while the displayed rotation itself gives $d_P = O(|t|)$. Hence $d_P = \Theta(|t|)$. No estimate with a power below two can hold on this path. □

6 Second variation in the configuration metric

The Frobenius normalization in Theorem 8 is different from the metric of signal configurations. We record the distinction explicitly. The calculation is a reformulation of local second variation, not a claim of priority for local optimality or the existence of a second-variation calculation.

We first complete the minimum-direction statement in the Gram metric. For a feasible sequence $A = R - I_m \to 0$, let $\zeta \geq 0$ be the Schur slack defined in Lemma 11. For $m \geq 4$ the sharp quotient in Theorem 8 is attained exactly when $\|A\mathbf{1}\|^2 + \zeta = o(\|A\|_F^2)$. Thus its nonzero limiting directions are all symmetric zero-diagonal, zero-row-sum matrices, with no excess second-order Schur slack. For $m = 3$ the corresponding condition is $\zeta = o(\|A\|_F^2)$: all nonzero critical tangent directions can attain the minimum. These conditions follow from the positive coefficients of $\|A\mathbf{1}\|^2$ and ζ in the expansions below, which specify the second-order feasibility condition as well as the first-order direction.

Fix a unit regular simplex $V = [v_1 \ \cdots \ v_m]$ in $\mathbb{R}^d$, $d \geq n = m-1$, and set $E = \operatorname{span}\{v_i\}$ and $r = d - n$. The configuration space is $(S^{d-1})^m$, with squared tangent norm $\|U\|_F^2 = \sum_i \|u_i\|^2$. At V the rotation directions are $U = \Omega V$, $\Omega^\mathsf{T} = -\Omega$. Their orthogonal complement will be called the horizontal tangent space. Write

$$\mathcal{H} = \{B = B^\mathsf{T} : \operatorname{diag} B = 0, \ \mathbf{1}^\mathsf{T} B \mathbf{1} = 0\}.$$

For $m \geq 3$ this space has the orthogonal decomposition

$$\mathcal{H} = \mathcal{H}_0 \oplus \mathcal{H}_1, \qquad \mathcal{H}_0 = \{B \in \mathcal{H} : B\mathbf{1} = 0\},$$

where $\mathcal{H}_1$ consists of the matrices $B_{ij} = a_i + a_j$ for $i \neq j$, $B_{ii} = 0$, and $a \perp \mathbf{1}$. The dimensions are $m(m-3)/2$ and $m-1$, respectively. For $m = 3$, $\mathcal{H}_0 = \{0\}$.

Theorem 10 (Horizontal second variation and its minimum)**.** *Fix a finite c, and retain k, h, q from Theorem 8; put $\sigma = 1 - qh > 0$. For $m \geq 3$ the horizontal tangent space is the orthogonal sum of the lifts of $\mathcal{H}_0$, the lifts of $\mathcal{H}_1$, and the normal-centroid space $\{z\mathbf{1}^\mathsf{T} : z \in E^\perp\}$. For a twice differentiable sphere path with horizontal initial velocity $U = U_0 + U_1 + U_\perp$,*

$$D_{R(t)}(c) = t^2 \left(k\alpha h^2 \|U_0\|_F^2 + \frac{k}{2}[h^2 + (m-2)\sigma]\,\|U_1\|_F^2 + \frac{k(m-1)}{2}\|U_\perp\|_F^2 \right) + o(t^2). \tag{27}$$

Thus the Hessian eigenvalues are twice the three displayed coefficients, with multiplicities $m(m-3)/2$, $m-1$, and r, omitting zero-dimensional spaces. On the full tangent space there is also the zero eigenvalue, of multiplicity $n(n-1)/2 + nr$, precisely the rotation directions.

The exact local coefficient of $D_R(c)$ relative to $d_P(X, \Delta)^2$ is the minimum of the coefficients in (27) *whose spaces are present. All minimum directions are precisely the direct sum of the spaces with that minimum coefficient, modulo rotations and the residual simplex symmetries. In particular:*

- *If $m \geq 4$ and $d = n$, the minimizing space is $\mathcal{H}_0$ when $h^2 < m\sigma$, $\mathcal{H}_1$ when $h^2 > m\sigma$, and their direct sum when equality holds.*
- *If $m \geq 4$ and $d \geq m$, the minimum is $k\alpha \min\{h^2, m/2\}$. The minimizing space is $\mathcal{H}_0$ when $h^2 < m/2$, the normal-centroid space when $h^2 > m/2$, and their direct sum when equality holds. The $\mathcal{H}_1$ coefficient is strictly larger than this minimum for every finite c.*
- *If $m = 3$ and $d = 2$, all nonzero horizontal directions minimize, with coefficient $k(1+ch)/2$. If $m = 3$ and $d \geq 3$, compare this coefficient to k: the two minimizing spaces are selected by $ch < 1$, $ch > 1$, or their direct sum at $ch = 1$.*

For $m = 2$ and $d \geq 2$, only the normal-centroid space remains, with coefficient $\phi(c)^2/2$ and horizontal multiplicity $d - 1$. For $m = 2$, $d = 1$, the configuration space is locally discrete and there is no nonconstant local variation.

Proof. Let $X(t)$ be a sphere path with $X(0) = V$, $X'(0) = U$, and $A(t) = \alpha(X(t)^\mathsf{T} X(t) - G_\Delta)$. Its initial Gram velocity is

$$B = A'(0) = \alpha(V^\mathsf{T} U + U^\mathsf{T} V) \in \mathcal{H}.$$

Since $S(t) = \alpha\|X(t)\mathbf{1}\|^2/2$ exactly,

$$S(t) = \frac{\alpha}{2} t^2 \|U\mathbf{1}\|^2 + o(t^2), \qquad B\mathbf{1} = \alpha V^\mathsf{T} U \mathbf{1}.$$

Using $VV^\mathsf{T} = \alpha^{-1} I_E$ and separating the normal projection U_N of U, equation (24) therefore gives

$$\frac{D_{R(t)}(c)}{kt^2} = \frac{h^2}{4}\|B\|_F^2 + \frac{\sigma}{2}\|B\mathbf{1}\|^2 + \frac{\alpha}{2}\|U_N\mathbf{1}\|^2 + o(1). \tag{28}$$

In particular the first variation vanishes and the second variation does not depend on acceleration.

Every $B \in \mathcal{H}$ has the unique horizontal in-plane lift

$$U_B = \frac{1}{2} VB + \frac{1}{2m} V(B\mathbf{1})\mathbf{1}^\mathsf{T}.$$

Direct multiplication proves its tangency, horizontality, and prescribed Gram velocity. Its squared norm is

$$\|U_B\|_F^2 = \frac{1}{4\alpha}\|B\|_F^2 + \frac{1}{2m\alpha}\|B\mathbf{1}\|^2. \tag{29}$$

The remaining horizontal vectors are exactly $z\mathbf{1}^\mathsf{T}$ with $z \in E^\perp$: all normal matrices with zero column sum are rotation directions. The two in-plane spaces are orthogonal both in Frobenius Gram norm and in (29). On $\mathcal{H}_0$ the metric is $\|B\|_F^2/(4\alpha)$. On $\mathcal{H}_1$,

$$\|B\mathbf{1}\|^2 = \frac{m-2}{2}\|B\|_F^2, \qquad \|U_B\|_F^2 = \frac{1}{2}\|B\|_F^2.$$

For $U_\perp = z\mathbf{1}^\mathsf{T}$ one has $\|U_\perp\mathbf{1}\|^2 = m\|U_\perp\|_F^2$. Substitution proves the spectrum. Every listed direction is attainable: use the independent spherical geodesics

$$x_i(t) = \cos(t\|u_i\|)v_i + \frac{\sin(t\|u_i\|)}{\|u_i\|}u_i,$$

with the continuous interpretation when $u_i = 0$. Furthermore $d_P(X(t), \Delta)^2 = t^2\|U\|_F^2 + o(t^2)$ for horizontal U. The path-free argument below justifies the corresponding local infimum over arbitrary sequences, rather than only over differentiable paths.

To verify the stated ordering when $m \geq 4$ and the normal space is present, use $h' = \sigma$ and $\sigma' = q(h^2 - \sigma)$. Put $f = \sigma - h^2/m$. At a zero of f,

$$f' = \frac{h}{m^2}(m(m-1) - (m+1)h^2) > 0 \quad \text{if } h^2 \leq m/2.$$

The elementary Gaussian-tail asymptotics give $\sigma/h^2 \to 1$ as $c \to -\infty$. Hence a first downward crossing is impossible and $\sigma > h^2/m$ whenever $h^2 \leq m/2$. At $h^2 = m/2$ this gives $\sigma > 1/2$. Subsequently σ cannot cross down through $1/2$, because $\sigma' = q(h^2 - 1/2) > 0$ on that boundary. These two strict inequalities imply that the $\mathcal{H}_1$ coefficient exceeds the smaller of the other two coefficients. The other comparisons follow by subtraction; for $m = 3$, $h^2 + \sigma = 1 + ch$.

For $m = 2$, write $A_{12} = \rho \geq 0$. Then $D_R(c) = \phi(c)^2\rho + O(\rho^2)$ and the exact Procrustes formula is $d_P^2 = 4(1 - \sqrt{1-\rho}) = 2\rho + O(\rho^2)$. □

Lemma 11 (Feasible-sequence expansion)**.** *Let $m \geq 3$ and let X be a unit-column configuration approaching the simplex orbit in $\mathbb{R}^d$, with $d \geq n$. Put $A = R - I_m$, $e = \mathbf{1}/\sqrt{m}$ and $C = PAP|_{e^\perp}$. Define*

$$\zeta = e^\mathsf{T}Ae - (PAe)^\mathsf{T}(I + C)^{-1}(PAe), \qquad B = A - \frac{2S}{m(m-1)}(J - I_m), \qquad b = B\mathbf{1}.$$

For $i \neq j$, set $(B_1)_{ij} = (b_i + b_j)/(m-2)$, set $(B_1)_{ii} = 0$ and put $B_0 = B - B_1$. Then $\zeta \geq 0$, $B_j \in \mathcal{H}_j$ and

$$E_0 = \frac{\|B_0\|_F^2}{4\alpha}, \qquad E_1 = \frac{\|B_1\|_F^2}{2}, \qquad E_\perp = \frac{\zeta}{\alpha}$$

satisfy

$$d_P^2 = E_0 + E_1 + E_\perp + o(d_P^2), \tag{30}$$

$$D_R(c) = k\alpha h^2 E_0 + \frac{k}{2}[h^2 + (m-2)\sigma]E_1 + \frac{k(m-1)}{2}E_\perp + o(d_P^2). \tag{31}$$

Here $\zeta = 0$ if and only if $\operatorname{rank} G = n$. *In particular $E_\perp = 0$ when $d = n$. The remainders apply to arbitrary feasible sequences.*

Proof. In the orthogonal splitting $\mathbb{R}e \oplus e^{\perp}$, the matrix $P + A$ has blocks

$$P + A = \begin{pmatrix} a & v^{\mathsf{T}} \\ v & I + C \end{pmatrix}, \qquad a = 2S/m, \quad v = PAe.$$

The lower block is positive definite near $A = 0$. The Schur complement therefore gives positivity exactly when $\zeta \geq 0$ and rank n exactly when $\zeta = 0$. Since $\|v\|^2 = \|A\mathbf{1}\|^2/m - 4S^2/m^2$, expansion of $(I + C)^{-1}$ gives

$$S = \frac{1}{2}\|A\mathbf{1}\|^2 + \frac{m}{2}\zeta + O(\|A\|_F^3 + S^2). \tag{32}$$

The exact Procrustes formula from Theorem 5 is

$$d_P^2 = 2m - \frac{2}{\alpha}\operatorname{tr}\sqrt{I + C}.$$

Since $\operatorname{tr} C = -2S/m$, its matrix square-root expansion gives

$$d_P^2 = \frac{2S}{m\alpha} + \frac{\|C\|_F^2}{4\alpha} + O(\|A\|_F^3).$$

The block decomposition also gives

$$\|C\|_F^2 = \|A\|_F^2 - \frac{2}{m}\|A\mathbf{1}\|^2 + \frac{4S^2}{m^2}.$$

Substitution of (32) yields

$$d_P^2 = \frac{\zeta}{\alpha} + \frac{\|A\|_F^2}{4\alpha} + \frac{\|A\mathbf{1}\|^2}{2m\alpha} + O(\|A\|_F^3 + S^2). \tag{33}$$

To control this remainder in the configuration metric, align V to a minimizer of d_P. Then $S = \alpha\|(X - V)\mathbf{1}\|^2/2 \leq \alpha m d_P^2/2$ and $\|A\|_F \leq 2\alpha\sqrt{m}\, d_P$. Thus the remainder is $o(d_P^2)$ uniformly for fixed m. Moreover $A - B = O(S)$, so replacing A by B in the quadratic terms changes them by $o(d_P^2)$. Orthogonality of B_0, B_1 and $\|B_1\mathbf{1}\|^2 = (m-2)\|B_1\|_F^2/2$ now prove (30). Substituting (32) into (24) gives

$$D_R(c)/k = \frac{h^2}{4}\|A\|_F^2 + \frac{\sigma}{2}\|A\mathbf{1}\|^2 + \frac{m}{2}\zeta + o(d_P^2).$$

The same orthogonal decomposition proves (31). □

Because the three energies are nonnegative, a feasible sequence attains the minimum quotient exactly when its higher-coefficient energies sum to $o(d_P^2)$. This proves the arbitrary-sequence claim in Theorem 10. The rotation-and-permutation quotient itself need not be smooth. Its tangent cone is $(\mathcal{H} \oplus E^{\perp})/(S_m \times O(r))$ with the horizontal metric. For $r > 0$, the normal-centroid component becomes a radial half-line in the quotient, while its horizontal multiplicity remains r.

The distinction between Gram and configuration metrics is explicit when $d \geq m$. For a unit normal $z \perp E$, set $x_i(t) = \sqrt{1 - t^2}\, v_i + tz$. Then

$$A = t^2(J - I_m), \qquad d_P^2 = 2m(1 - \sqrt{1 - t^2}) = mt^2 + O(t^4), \qquad \Big\| \sum_i x_i(t) \Big\|^2 = m^2 t^2.$$

The squared Gram defect has order t^4, whereas the squared configuration distance has order t^2. The separate centroid term controls this transverse path.

Corollary 12 (Decoding and mean-width second variations)**.** *Let $\mu = \lambda/\sqrt{\alpha}$ and define*

$$\omega_\lambda(c) = \frac{\mu e^{-\mu^2/2}}{m} e^{\mu c}\phi(c)^2\Phi(c)^{m-2}, \qquad A_\lambda = \int_{\mathbb{R}} \omega_\lambda h^2, \quad B_\lambda = \int_{\mathbb{R}} \omega_\lambda \sigma, \quad C_\lambda = \int_{\mathbb{R}} \omega_\lambda.$$

All three constants are finite and positive. The quadratic coefficients of the decoding deficit on the three horizontal spaces are exactly

$$\Lambda_0 = \alpha A_\lambda, \qquad \Lambda_1 = \frac{A_\lambda + (m-2)B_\lambda}{2}, \qquad \Lambda_\perp = \frac{m-1}{2} C_\lambda.$$

The local coefficient relative to d_P^2 is their minimum over spaces present, and every minimum direction belongs to the direct sum of the spaces attaining that minimum. Theorems 13 and 14 below determine the ordering of these decoding coefficients at every positive channel parameter.

Relative to the Gram Frobenius norm, the exact decoding coefficient is

$$\liminf_{G\to G_\Delta,\ G\neq G_\Delta} \frac{\delta_\lambda(G)}{\|G - G_\Delta\|_F^2} = \begin{cases} \alpha^2(A_\lambda + B_\lambda)/4, & m = 3, \\ \alpha^2 A_\lambda/4, & m \geq 4. \end{cases}$$

These coefficients are attainable in dimension n. For $m = 2$,

$$\frac{\delta_\lambda(G)}{\|G - G_\Delta\|_F} \longrightarrow \frac{\lambda\phi(\lambda)}{4\sqrt{2}}, \qquad \frac{\delta_\lambda(G)}{d_P^2} \longrightarrow \frac{\lambda\phi(\lambda)}{4}$$

along nonconstant configurations in dimension at least two.

For $m \geq 3$, the same assertions for the unnormalized expected-maximum deficit hold after replacing $\omega_\lambda(c)$ by $\omega_w(c) = \phi(c)^2\Phi(c)^{m-2}/\sqrt{\alpha}$. For normalized spherical mean width, multiply all coefficients by $2/\mathbb{E}\|g\|$, $g \sim N(0, I_d)$. For $m = 2$ in dimension at least two, the expected-maximum deficit has local coefficients $\phi(0)/(2\sqrt{2})$ relative to $\|G - G_\Delta\|_F$ and $\phi(0)/2$ relative to d_P^2.

Proof. Use the integral identities already established in the manuscript. Near $R = I_m$, covariance-density derivatives through order three are bounded by a polynomial times a fixed nondegenerate Gaussian density. Their products with $e^{\mu \max_i x_i}$, or with $\max_i x_i$, remain integrable. Thus differentiating the expectations twice is justified, and their Taylor coefficients equal the displayed integrals. Equivalently, differentiate the distribution-function identities with a common integrable domination: for negative c bound the fixed orthant integral by a Gaussian tail, and for positive c use that each derivative density has total integral zero and integrate over the complementary tail. This also proves convergence of the coefficient integrals.

Integrating the path-free expansion in Lemma 11, or applying this density Taylor expansion directly, proves the coordinate coefficients and their minimum criterion. For the Gram coefficients, the expansion is

$$\delta_\lambda(G) = \frac{A_\lambda}{4}\|A\|_F^2 + \frac{B_\lambda}{2}\|A\mathbf{1}\|^2 + \frac{mC_\lambda}{2}\zeta + o(\|A\|_F^2)$$

on sequences with $S = O(\|A\|_F^2)$. Sequences for which $S/\|A\|_F^2 \to \infty$ have divergent deficit quotient, by the positive centroid bound already proved. For $m \geq 4$ the lower coefficient is attained by a nonzero row-zero Gram direction with zero Schur slack. For $m = 3$ use $\|A\mathbf{1}\|^2 = \|A\|_F^2/2 + S^2$ and a rank-two feasible path. Finally $\|A\|_F = \alpha\|G - G_\Delta\|_F$. For $m = 2$ the exact success probability is $\Phi(\lambda\sqrt{1-\rho})$, $\rho = A_{12}$; the stated constants follow by differentiation and the Procrustes formula above. □

7 Unique channel transitions and exact small-shell minimizers

Corollary 12 expresses the directional coefficients through three explicit integrals. We now determine the ordering of these coefficients for every positive channel parameter and, when the normal coefficient is strictly least, identify the exact optimizer on every sufficiently small Procrustes shell. The radii obtained here are uniform in the ambient dimension, but are not evaluated numerically.

7.1 A tilted Gaussian family

For $m \geq 3$ and $\mu \in \mathbb{R}$, set

$$\begin{aligned} w_{m,\mu}(c) &= e^{\mu c}\phi(c)^2\Phi(c)^{m-2}, & Z_m(\mu) &= \int_{\mathbb{R}} w_{m,\mu}(c)\,dc, \\ a_m(\mu) &= Z_m(\mu)^{-1}\int_{\mathbb{R}} h(c)^2 w_{m,\mu}(c)\,dc, & b_m(\mu) &= Z_m(\mu)^{-1}\int_{\mathbb{R}} \sigma(c) w_{m,\mu}(c)\,dc, \\ H_m(\mu) &= Z_m(\mu)^{-1}\int_{\mathbb{R}} h(c) w_{m,\mu}(c)\,dc. \end{aligned} \tag{34}$$

Here $q = \phi/\Phi$, $h(c) = c + q(c)$ and $\sigma = 1 - qh$, as in Theorem 10. The normalizer is positive and finite. At $\mu = \lambda/\sqrt{\alpha}$,

$$A_\lambda = C_\lambda a_m(\mu), \qquad B_\lambda = C_\lambda b_m(\mu), \qquad C_\lambda = \frac{\mu e^{-\mu^2/2}}{m} Z_m(\mu) > 0.$$

Consequently the normalized decoding curvatures are

$$L_0 = \frac{m-1}{m} a_m(\mu), \qquad L_1 = \frac{a_m(\mu) + (m-2)b_m(\mu)}{2}, \qquad L_\perp = \frac{m-1}{2}. \tag{35}$$

The endpoint $\mu = 0$ below is an endpoint of the normalized moment family; it is not division by C_λ at a zero-amplitude channel.

Theorem 13 (Unique ordered channel crossings)**.** *Let $m \geq 4$. There are unique positive parameters*

$$0 < \mu_{1\perp}(m) < \mu_{0\perp}(m) < \mu_{01}(m)$$

satisfying, respectively,

$$a_m + (m-2)b_m = m - 1, \qquad a_m = m/2, \qquad a_m = m b_m.$$

Each equation has no other root in $[0, \infty)$. For $\lambda > 0$, with $\mu = \lambda/\sqrt{\alpha}$, the four open intervals have the orderings

$$\begin{aligned} 0 < \mu < \mu_{1\perp}: &\quad \Lambda_0 < \Lambda_1 < \Lambda_\perp, \\ \mu_{1\perp} < \mu < \mu_{0\perp}: &\quad \Lambda_0 < \Lambda_\perp < \Lambda_1, \\ \mu_{0\perp} < \mu < \mu_{01}: &\quad \Lambda_\perp < \Lambda_0 < \Lambda_1, \\ \mu > \mu_{01}: &\quad \Lambda_\perp < \Lambda_1 < \Lambda_0. \end{aligned}$$

At the three endpoints the indicated pair agrees, and the remaining inequality is strict. In particular $\min\{\Lambda_0, \Lambda_\perp\} < \Lambda_1$ for every $\lambda > 0$. The thresholds in λ are $\sqrt{\alpha}$ times these thresholds.

Proof. Appendix A proves continuity and strict increase of a_m, b_m, H_m and a_m/b_m, as well as

$$\frac{1}{2} < b_m(\mu) < 1 \quad (m \geq 4,\ \mu \geq 0), \qquad 2a_m + (m-1)b_m = m + \mu H_m.$$

At zero the identity gives $a_m < m/2$ and, using $b_m < 1$, $a_m + (m-2)b_m < m-1$. Using $b_m > 1/2$ also gives $a_m < mb_m$. Moreover $2a_m(\mu) > 1 + \mu H_m(0)$, so a_m tends to infinity. The three continuous strictly increasing functions $a_m + (m-2)b_m$, a_m, and a_m/b_m therefore cross their targets once. At $a_m = m/2$, the bound $b_m > 1/2$ puts the first crossing to the left and the third to the right. Finally $L_1 - L_0 = (m-2)(mb_m - a_m)/(2m)$ gives the displayed sign chart. The analytic ingredients, including the strict benchmark at $m = 4$, are proved in full in Appendix A. □

Theorem 14 (The triangular channel crossing)**.** *For $m = 3$ there is a unique $\mu_*(3) > 0$ such that $\mu_* H_3(\mu_*) = 1$, equivalently $a_3(\mu_*) + b_3(\mu_*) = 2$. For $0 < \mu < \mu_*$ one has $\Lambda_1 < \Lambda_\perp$; for $\mu > \mu_*$ the reverse inequality holds. They agree only at μ_*. The space $\mathcal{H}_0$ is absent.*

Proof. The balance identity gives $(a_3 + b_3)/2 = (3 + \mu H_3)/4$. On $[0, \infty)$, the function $\mu H_3(\mu)$ is continuous and strictly increasing from zero to infinity, since H_3 is positive and increasing. This proves the assertion. Its integral ingredients are included in Appendix A. □

7.2 The exact strict-normal shell optimum

For $m \geq 3$ and $\lambda > 0$, impose the strict-normal condition

$$\begin{cases} \mu > \mu_{0\perp}(m), & m \geq 4, \\ \mu > \mu_*(3), & m = 3, \end{cases} \qquad \mu = \lambda/\sqrt{\alpha}. \tag{36}$$

Equivalently $a_m(\mu) > m/2$ for $m \geq 4$, and $a_3(\mu) + b_3(\mu) > 2$ for $m = 3$. These are precisely the conditions that the normal coefficient be strictly smaller than every in-plane coefficient whose space is present.

Theorem 15 (Exact optimizer on small Procrustes shells)**.** *Assume* (36)*. There is $\rho_0 = \rho_0(m, \lambda) > 0$ such that, for every $d \geq m$, every unit regular simplex $V = [v_1 \ \cdots \ v_m]$ in $\mathbb{R}^d$, and every $0 < \rho < \rho_0$, the minimum of $\delta_\lambda(X) = P_{\mathrm{c}}(G_\Delta; \lambda) - P_{\mathrm{c}}(G_X; \lambda)$ over all unit configurations with $d_P(X, V) = \rho$ is attained precisely by the orthogonal images of*

$$N_i = \kappa v_i + \sqrt{1 - \kappa^2}\, z, \qquad \kappa = 1 - \frac{\rho^2}{2m}, \qquad \|z\| = 1, \quad z \perp \operatorname{span}\{v_i\}. \tag{37}$$

The normal z may be chosen once for all these radii. Different choices of z give the same orthogonal orbit. The minimal deficit is exactly

$$\delta_{\min}(\rho) = P_{\mathrm{c}}(G_\Delta; \lambda) - P_{\mathrm{c}}\left(G_\Delta; \lambda\left(1 - \frac{\rho^2}{2m}\right)\right). \tag{38}$$

Proof. Appendix B supplies the complete geometric argument. The centered square root $T = (\alpha P G_X P)^{1/2}$ satisfies, on a uniform neighborhood, $d_P(X, V)^2 = 2m - 2\operatorname{tr}(T)/\alpha$. On the shell $\tau = \rho^2$, write $T = \kappa P + D$, where D is symmetric, centered and trace-free. The covariance reconstructed from (τ, D) defines a smooth extension of the decoding deficit. Permutation symmetry makes $D = 0$ stationary for every small τ. Its Hessian at $(0, 0)$ is positive definite under (36); the precise coefficients and the square-root coordinate normalization are derived in Appendix B. Continuity then gives a strict quadratic lower bound near this ray. Every sufficiently small actual shell lies in that neighborhood. Thus $D = 0$ is its unique minimizing covariance. It is realized by (37), with exact distance ρ, and equality of Gram matrices is

equivalent to an orthogonal change of realization. A common Gaussian shift in the normal direction gives (38). The neighborhood estimates are made in the fixed $m \times m$ Gram space, so ρ_0 is independent of d and V. □

Theorem 16 (Quartic expansion with a sixth-order remainder)**.** *Let $p(\lambda) = P_c(G_\Delta; \lambda)$. This function is real analytic. For fixed $m \geq 3$ and $\lambda > 0$ satisfying* (36)*, there are $C = C(m, \lambda) > 0$ and $r = r(m, \lambda) > 0$ such that Theorem 15 holds for $0 < \rho < r$ and*

$$\delta_{\min}(\rho) = \frac{\lambda p'(\lambda)}{2m}\rho^2 - \frac{\lambda^2 p''(\lambda)}{8m^2}\rho^4 + \mathcal{R}_{m,\lambda}(\rho), \tag{39}$$

$$|\mathcal{R}_{m,\lambda}(\rho)| \leq C|\rho|^6 \qquad (|\rho| < r). \tag{40}$$

For negative ρ, the deficit in this display denotes the even extension given by the right side of (38)*. All its odd derivatives at zero vanish. In particular there is no cubic term. The leading coefficient satisfies*

$$\frac{\lambda p'(\lambda)}{2m} = \Lambda_\perp = \frac{m-1}{2}C_\lambda. \tag{41}$$

Proof. All real exponential moments of the Gaussian maximum in (2) are finite, so its moment generating function, and hence p, is real analytic. Taylor's theorem through the quadratic term at λ has error $O(|s-\lambda|^3)$. Substituting $s = \lambda - \lambda\rho^2/(2m)$ gives (39)–(40); the sign of the quartic term comes from subtracting $p(s)$ from $p(\lambda)$. The exact expression is even in ρ, which gives the assertion about all odd derivatives. The normal covariance path has initial velocity $(J - I_m)/m$ in the squared radius, so its first deficit derivative is $C_\lambda(m-1)/2$. Differentiating the exact value gives the other side of (41). Detailed domination and remainder arguments are in Appendix B. □

The equal-curvature boundary is excluded, and no claim of a finite-radius in-plane optimizer is made. The positive radius and remainder constant are existential; they do not supply an effective numerical certificate. The channel crossings, the strict-normal optimization theorem and the quantified Taylor expansion are formalized over the actual Gaussian integrals and orthogonal Procrustes distance. Appendix C records the exact scope, theorem correspondence and reproducible Lean checks; the other global bounds and numerical experiments in this paper are not included in that formalization.

8 Finite-dimensional decoding diagnostics

The local coefficients describe directional sensitivity near a simplex. We test that use separately from fixed-budget learning and from the numerical size of the global recovery constant. All comparisons use the channel in (2), uniform messages and unit-energy codewords.

8.1 Directional prediction

We use $(m, d) = (4, 3)$ and $(16, 15)$, with $\lambda \in \{1, 2, 3\}$. For each of the two in-plane sectors, three independently generated tangent directions have unit configuration Frobenius norm. Spherical geodesics preserve each codeword's energy. At amplitudes $t \in \{0.05, 0.10, 0.20\}$, the target is the central average $[\delta_\lambda(X(t)) + \delta_\lambda(X(-t))]/2$. The predictor is $\Lambda_j t^2$ for a direction in sector j. Two scalar controls replace Λ_j by its dimension-weighted average or by the equally weighted average of the two sectors. These controls measure the effect of discarding direction information.

The resulting 108 cells use eight independently randomized Sobol integrations, with 2^{11} points per replicate. Table 1 reports mean absolute prediction error over 18 cells per row. The scalar controls have the same displayed aggregate error on this balanced grid, although their

cellwise predictions differ. The directional predictor has smaller aggregate error in every row. This comparison does not establish a uniform cellwise ordering. For one four-signal cell at $\lambda = 3$ and $t = 0.20$, its error is 2.54×10^{-5}, compared with 1.99×10^{-5} for the equal-sector control.

Table 1: Mean absolute error for the central decoding-loss prediction. Each row aggregates both in-plane sectors, three directions and three channel parameters.

m	t	Directional	Dimension-weighted	Equal-sector
4	0.05	3.89×10^{-8}	3.25×10^{-6}	3.25×10^{-6}
4	0.10	6.68×10^{-7}	1.31×10^{-5}	1.31×10^{-5}
4	0.20	1.09×10^{-5}	5.43×10^{-5}	5.43×10^{-5}
16	0.05	4.47×10^{-8}	1.08×10^{-5}	1.08×10^{-5}
16	0.10	1.88×10^{-7}	4.31×10^{-5}	4.31×10^{-5}
16	0.20	1.01×10^{-6}	1.73×10^{-4}	1.73×10^{-4}

Central averaging removes odd-order terms and changes the prediction target. For a four-signal direction at $\lambda = 2$ and $t = 0.20$, the two losses are 0.0026961 and 0.0023393. Their average is 0.0025177, while the quadratic prediction is 0.0025202. A one-sided tolerance requires a separate remainder bound. All 108 losses exceed five estimated integration standard errors. That check resolves the loss, rather than the much smaller prediction residual. Eight prespecified higher-resolution checks have paired difference intervals containing zero. Their differences can still exceed the residuals in Table 1. At $m = 16$ and $\lambda = 2$, one checked direction changes by -1.48×10^{-7} at $t = 0.05$ and by -2.37×10^{-6} at $t = 0.20$. The table therefore supports a coarse directional diagnostic, without a certified residual, improvement factor or higher-order coefficient.

8.2 A fixed-budget learning comparison

We adapt the communication autoencoder of O'Shea and Hoydis [13, Section III-A and Table IV] to the two dimensions above. The encoder uses a dense layer of width m with rectified linear activation, a linear layer of width d and per-codeword energy normalization. The decoder uses a dense layer of width m with rectified linear activation and a softmax output of width m. Training minimizes cross-entropy at $E_b/N_0 = 7$ dB, where $E_b/N_0 = \lambda^2/(2 \log_2 m)$. An additional arm uses the ordinary simplex Gram mean-square penalty with weight 0.1. This penalty follows from the known optimal geometry. The local decoding Hessian is not a cross-entropy Hessian or an Adam preconditioner.

Five paired initializations per dimension use identical message and noise streams. Both arms use double precision and Adam with learning rate 0.001 for 1200 steps. The batch size is 256 for the first 600 steps and 1024 for the remaining 600 steps. Every run contributes its final checkpoint, with no seed or checkpoint selection. These 20 runs complete a fixed budget, without an optimization-convergence claim. The dimensions differ from the original autoencoder's $(16, 7)$ example.

Table 2 evaluates each learned codebook with matched maximum-likelihood decoding. Block error rate is one minus decoding success probability. Each interval is a paired Student-t interval over five training initializations, evaluated pointwise without multiplicity correction. The eight mean differences favor the Gram penalty under this budget. Both arms retain a gap from the ideal simplex. The learned decoder is a different endpoint. In two of the five 16-signal pairs, its estimated error increases at all four tested channel parameters. Cross-initialization paired intervals for that decoder were not evaluated, so the comparison establishes no end-to-end decoder improvement.

Table 2: Mean matched-decoder block error rate over five paired initializations. The interval is for the Gram-penalized arm minus the baseline.

m	λ	Baseline	Gram penalty	Simplex	Paired 95% interval
4	1	0.421808	0.403936	0.399550	$[-0.032181, -0.003562]$
4	2	0.151550	0.127305	0.121023	$[-0.043128, -0.005362]$
4	3	0.034736	0.022434	0.019231	$[-0.021918, -0.002685]$
4	4.47744	0.001919	0.000667	0.000375	$[-0.002372, -0.000131]$
16	1	0.749215	0.746866	0.729857	$[-0.003146, -0.001552]$
16	2	0.434459	0.428081	0.383069	$[-0.008566, -0.004190]$
16	3	0.165418	0.158813	0.116521	$[-0.008638, -0.004573]$
16	6.33206	0.0003243	0.0002622	0.0000273	$[-0.0000924, -0.0000318]$

Evaluation retains independent direct Gaussian sampling with 100000 trials per codebook and channel parameter. The integration evaluation uses eight randomized Sobol replicates with 2^{12} points each. Training variability and integration variability remain distinct sources of uncertainty. The computations used four CPU workers under Linux, Python 3.11.16, NumPy 2.2.0, SciPy 1.17.1, PyTorch 2.8.0 and mpmath 1.3.0. No timing comparison is made.

8.3 Size of the global recovery constant

The explicit constant in (15) is positive but very small. Table 3 evaluates a conservative partition envelope for its scalar maximization. Repeating the $\lambda = 2$ cases at 80 and 120 decimal digits gives consistent values. Such agreement is not outward-rounded interval certification.

Table 3: Numerical values of the explicit global Procrustes constant c_P.

m	$\lambda = 1$	$\lambda = 2$	$\lambda = 3$
4	2.1293×10^{-39}	1.2026×10^{-39}	1.4640×10^{-40}
16	4.6152×10^{-233}	3.5491×10^{-233}	7.5743×10^{-234}

At $\lambda = 2$, certifying $d_P \leq 0.1$ through this inverse bound would require a reliable deficit upper bound of 1.20×10^{-41} for four signals, or 3.55×10^{-235} for 16 signals. These values are far below the precision of the empirical estimates. The explicit global bound supplies a rate, while practical geometric recovery requires sharper constants and an effective numerical lower bound for the local transition radius.

9 Conclusion and limitations

The explicit comparison controls Gram defect and centroid displacement, including singular configurations. Together they give a global quadratic configuration bound at fixed m and positive λ. The local expansion identifies the feasible least-loss sectors in the configuration metric. Numerical diagnostics support its aggregate use for small central perturbations of four- and 16-signal constellations. The normalized Gaussian moment family has three unique ordered crossings for $m \geq 4$ and a separate unique crossing for $m = 3$. In the strict-normal regime the finite-radius small-shell optimizer is exactly the common-normal deformation, up to orthogonal equivalence. Its deficit has a quartic expansion with a quantified $O(\rho^6)$ remainder and no odd Taylor terms. The radius and remainder constant depend only on m and λ, not on the embedding dimension. The global comparison, recovery and mean-width results, together with the transition, strict-normal optimization and remainder results, have been checked in Lean

over the actual Gaussian integrals. The global constants are included; numerical diagnostics and experimental claims remain outside the certificate.

The global constants deteriorate rapidly with the number of signals. Their numerical values provide no practical geometry certificate in the tested cases. Practical recovery needs sharper constants, an effective numerical lower bound for the local transition radius, computable remainder bounds, and controlled integration error. The fixed-budget learning comparison concerns known Gram regularization and matched decoding. Its decoder and convergence limitations remain separate from the mathematical stability result. The rank penalty quantifies a gap from the unrestricted simplex value, without identifying the optimal rank-constrained constellation. The channel model assumes equal message probabilities and isotropic Gaussian noise. Fading, synchronization, channel estimation and constrained modulation alphabets require separate analysis.

A Proof of the channel transitions

This appendix proves the scalar and integral assertions underlying Theorems 13 and 14. Fix an integer $m \geq 3$ and put $\alpha = (m-1)/m$. For $c \in \mathbb{R}$, let

$$\phi(c) = (2\pi)^{-1/2} e^{-c^2/2}, \qquad \Phi(c) = \int_{-\infty}^{c} \phi(s)\, ds,$$
$$q(c) = \frac{\phi(c)}{\Phi(c)}, \qquad h(c) = c + q(c), \qquad \sigma(c) = 1 - q(c)h(c).$$

These functions are smooth because $0 < \Phi(c) < 1$. For $\mu \in \mathbb{R}$, define

$$w_{m,\mu}(c) = e^{\mu c}\phi(c)^2\Phi(c)^{m-2}, \qquad Z_m(\mu) = \int_{\mathbb{R}} w_{m,\mu}(c)\, dc, \qquad p_{m,\mu}(c) = \frac{w_{m,\mu}(c)}{Z_m(\mu)}. \tag{A.1}$$

The density is positive and has Gaussian tails; in particular $0 < Z_m(\mu) < \infty$. Write

$$a_m(\mu) = \mathbb{E}_{m,\mu}[h^2], \qquad b_m(\mu) = \mathbb{E}_{m,\mu}[\sigma], \qquad H_m(\mu) = \mathbb{E}_{m,\mu}[h].$$

We omit m when it is fixed. The channel parameter of Section 7 is $\lambda > 0$, and $\mu = \lambda/\sqrt{\alpha}$. The endpoint $\mu = 0$ is used only for the continuous family of these normalized moments. Define

$$L_0(m,a) = \frac{(m-1)a}{m}, \qquad L_1(m,a,b) = \frac{a + (m-2)b}{2}, \qquad L_\perp(m) = \frac{m-1}{2}. \tag{A.2}$$

Their relation to the unnormalized decoding curvatures is recorded in Subsection A.9.

A.1 Scalar differential identities

Lemma A.1. *$q' = -qh$, $h' = \sigma$, and $\sigma' = q(h^2 - \sigma)$.*

Proof. Since $\phi' = -c\phi$,

$$q' = \frac{\phi'\Phi - \phi^2}{\Phi^2} = -q\Big(c + \frac{\phi}{\Phi}\Big) = -qh.$$

Then $h' = 1 + q' = 1 - qh = \sigma$, and

$$\sigma' = -q'h - qh' = qh \cdot h - q\sigma = q(h^2 - \sigma).$$

□

Lemma A.2. *For every real c, one has $h(c) > 0$ and $0 < \sigma(c) < 1$. In particular h and h^2 are strictly increasing, and q is strictly decreasing. Also*

$$0 < h(c) \le \max(c, 0) + \sqrt{2/\pi},$$

with equality on the right at $c = 0$.

Proof. Let g be standard normal and let Y be the law of $c + g$ conditioned on $c + g > 0$. The conditioning event has probability $\Phi(c) \in (0, 1)$, and the conditional law is supported on $(0, \infty)$ with a non-degenerate absolutely continuous distribution, so its mean is positive and its variance is positive and finite. Write $Z = c+g$, so Z has density $\phi(z-c)$ and $P(Z > 0) = \Phi(c)$. The substitution $u = z - c$ and the identity $(\phi)' = -u\phi$ give

$$\mathbb{E}[Z\mathbf{1}_{Z>0}] = \int_{-c}^{\infty} (u + c)\phi(u)\, du = \phi(c) + c\Phi(c),$$

because $\int_{-c}^{\infty} u\phi(u)\, du = \phi(c)$. Dividing by $\Phi(c)$ produces $\mathbb{E}[Y] = c + q(c) = h(c)$. For the second moment, $u^2\phi(u) = \phi(u) - (u\phi(u))'$ and $[u\phi(u)]_{-c}^{\infty} = c\phi(c)$, so $\int_{-c}^{\infty} u^2\phi = \Phi(c) - c\phi(c)$. Adding $\int_{-c}^{\infty}(2cu + c^2)\phi = 2c\phi(c) + c^2\Phi(c)$ yields

$$\mathbb{E}[Z^2\mathbf{1}_{Z>0}] = \Phi(c)(1 + c^2) + c\phi(c),$$

hence $\mathbb{E}[Y^2] = 1 + c^2 + cq(c) = 1 + ch(c)$. Therefore $\mathrm{Var}(Y) = 1 + ch - h^2$. The identity $h = c + q$ turns the right-hand side into $1 - qh = \sigma$. Thus $h > 0$ and $\sigma > 0$. Since $q > 0$ and $h > 0$, one has $qh > 0$, so $\sigma < 1$.

Now $h' = \sigma > 0$, so h is strictly increasing, and $(h^2)' = 2h\sigma > 0$. Also $q' = -qh < 0$, so q is strictly decreasing. Therefore $h(c) < h(0)$ for $c < 0$ and $q(c) \le q(0)$ for $c \ge 0$. The value $h(0) = q(0) = \phi(0)/\Phi(0) = \sqrt{2/\pi}$ gives the stated bound.

□

A.2 The zero-locus derivatives

Define the scalar functions

$$F(c) = h(c)^2 - \sigma(c), \qquad D(c) = \sigma + \sigma^2 - (1 - \sigma)h^2.$$

Both are smooth. The chain rule and Lemma A.1 give, at every c,

$$F' = 2h\sigma - qF.$$

In particular, on the locus $F = 0$,

$$F' = 2h\sigma > 0,$$

because $h > 0$ and $\sigma > 0$.

For D, use $q = (1 - \sigma)/h$, which is the rearrangement of $\sigma = 1 - qh$. Then

$$\sigma' = \frac{(1 - \sigma)(h^2 - \sigma)}{h}, \qquad (h^2)' = 2h\sigma.$$

Writing $s = \sigma(c)$ and $x = h(c)^2$ at a fixed c,

$$D'(c) = \frac{(1 - s)(x - s)}{h}\,(1 + 2s + x) - 2hs(1 - s). \tag{A.3}$$

This is an identity of smooth functions, obtained by the chain rule, not by differentiating a remainder.

Lemma A.3 (positivity at a zero)**.** *Let $h, s \in \mathbb{R}$ satisfy $0 < h$, $0 < s < 1$, and*

$$s + s^2 - (1 - s)h^2 = 0.$$

Then $x = h^2$ equals $s(1 + s)/(1 - s)$, and the right-hand side of (A.3) *equals*

$$\frac{4s^3}{h(1 - s)} > 0.$$

Thus, whenever $D(c) = 0$, one has $D'(c) > 0$.

Proof. The constraint solves as $x = s(1 + s)/(1 - s)$. Then

$$x - s = \frac{2s^2}{1 - s}, \qquad \sigma' \text{ coefficient} = \frac{1 - s}{h} \cdot \frac{2s^2}{1 - s} = \frac{2s^2}{h}.$$

Substitute x into $1 + 2s + x$:

$$1 + 2s + x = \frac{(1 + 2s)(1 - s) + s(1 + s)}{1 - s} = \frac{1 + 2s - s^2}{1 - s}.$$

The expression (A.3) becomes

$$\begin{aligned} &\frac{2s^2}{h} \cdot \frac{1 + 2s - s^2}{1 - s} - 2hs(1 - s) \\ &\qquad = \frac{2s^2(1 + 2s - s^2) - 2h^2 s(1 - s)^2}{h(1 - s)}. \end{aligned}$$

The relation $h^2 = s(1 + s)/(1 - s)$ gives $h^2(1 - s)^2 = s(1 + s)(1 - s) = s(1 - s^2)$, so the second numerator term is $2s^2(1 - s^2)$. The numerator collapses to

$$2s^2\big(1 + 2s - s^2 - 1 + s^2\big) = 4s^3.$$

Hence the expression equals $4s^3/(h(1 - s))$. The hypotheses $h > 0$ and $0 < s < 1$ make this positive.

□

The same computation with no constraint shows the sign identity used later for the ratio:

$$\Big(\frac{h^2}{\sigma}\Big)' = \frac{h\,D}{\sigma^2}.$$

Indeed, with $x = h^2$ and $s = \sigma$,

$$\Big(\frac{x}{s}\Big)' = \frac{x's - xs'}{s^2}, \qquad x' = 2hs, \qquad s' = \frac{(1 - s)(x - s)}{h},$$

and clearing the positive factor h produces the numerator hD if and only if

$$2s^2 - (1 - s)(x - s) = s + s^2 - (1 - s)x,$$

which is an algebraic identity. Thus, wherever $D > 0$, the ratio h^2/σ is strictly increasing. The sign of D is proved in Subsection A.4, after the tail expansion.

A.3 Mills expansion with a bounded remainder

For $t > 0$ the left-tail Mills ratio is

$$M(t) = \frac{\Phi(-t)}{\phi(t)} = \int_0^\infty \exp(-tv - v^2/2)\, dv.$$

The substitution $s = tv$ gives the Laplace representation,

$$M(t) = t^{-1} \int_0^\infty e^{-s} \exp(-s^2/(2t^2))\, ds.$$

At $c = -t$ one has $q(-t) = 1/M(t)$ and $h(-t) = 1/M(t) - t$.

Lemma A.4 (explicit Taylor remainder)**.** *Let $n \geq 1$ be an integer and let $t > 0$. Then*

$$M(t) = \sum_{k=0}^{n-1} (-1)^k \frac{(2k)!}{k!\, 2^k}\, t^{-(2k+1)} + \theta_n(t),$$

where

$$|\theta_n(t)| \leq \frac{(2n)!}{n!\, 2^n\, t^{2n+1}}.$$

In particular, for $n = 5$,

$$M(t) = t^{-1} - t^{-3} + 3t^{-5} - 15t^{-7} + 105t^{-9} + \theta_5(t),$$

with $|\theta_5(t)| \leq 945\, t^{-11}$. The remainder is not differentiated anywhere below.

Proof. For $x > 0$ the Lagrange remainder after the terms of degree $0, \ldots, n-1$ in e^{-x} has absolute value at most $x^n/n!$, because it equals $(-1)^n e^{-\xi} x^n/n!$ for some $\xi \in (0, x)$ and $e^{-\xi} \leq 1$; at $x = 0$ the remainder is zero. Insert $x = s^2/(2t^2)$, integrate against e^{-s} on $(0, \infty)$, and use $\int_0^\infty e^{-s} s^{2n}\, ds = (2n)!$. Dividing by t gives the bound on θ_n. The coefficient of $t^{-(2k+1)}$ is $(-1)^k (2k)!/(k! 2^k)$. For $k \leq 4$ these coefficients are $1, -1, 3, -15, 105$, and the $n = 5$ bound is $10!/(120 \cdot 32) = 945$.

□

Set $u = t^{-2}$. For every $t > 0$,

$$M(t) = t^{-1}\big(1 - u + 3u^2 - 15u^3 + 105u^4 + r(t)\big), \tag{A.4}$$

where $|r(t)| \leq 945\, u^5$. Write $a_0(u) = -u + 3u^2 - 15u^3 + 105u^4$ and $a(t) = a_0(u) + r(t)$, so the factor in parentheses is $1 + a(t)$.

Lemma A.5 (algebraic inversion on a far ray)**.** *There exist a finite $T_0 \geq 1$ and a finite positive constant C_h such that for every $t \geq T_0$,*

$$h(-t) = t^{-1} - 2t^{-3} + 10t^{-5} - 74t^{-7} + \rho_h(t), \qquad |\rho_h(t)| \leq C_h\, t^{-9}. \tag{A.5}$$

The coefficient -74 is required later. Stopping the expansion of h at $O(t^{-7})$ does not determine the coefficient of t^{-6} in σ.

Proof. As $u \to 0$, $a_0(u) \to 0$ and $r(t) \to 0$. Choose a finite $T_0 \geq 1$ such that $t \geq T_0$ implies both $|a_0(u)| \leq 1/4$ and $|r(t)| \leq 1/4$. Then $|a(t)| \leq 1/2$ and $|a_0(u)| \leq 1/2$, so $1 + a(t) \geq 1/2$ and $q(-t) = t/(1 + a(t))$ is well-defined. In particular

$$h(-t) = t\Big(\frac{1}{1 + a(t)} - 1\Big).$$

On the compact interval $0 \le u \le T_0^{-2}$ one has $|a_0(u)| \le 1/2$, so $g(u) = (1+a_0(u))^{-1}$ is smooth there. The recurrence below computes the Taylor coefficients of g at 0, because it is the Cauchy product for the reciprocal of the polynomial $1 + a_0$. Lagrange's theorem in the variable u, through degree 4, then gives a remainder bounded by a constant times u^5: the fifth derivative of g is continuous on that compact interval. The recurrence $b_0 = 1$ and $b_n = -\sum_{j=1}^{n} a_j b_{n-j}$ for the coefficients of $(1 + a_0)^{-1}$, with $a_1 = -1$, $a_2 = 3$, $a_3 = -15$, $a_4 = 105$, gives

$$\begin{aligned} b_1 &= 1, \\ b_2 &= -(-1+3) = -2, \\ b_3 &= -(2+3-15) = 10, \\ b_4 &= -(-10-6-15+105) = -74. \end{aligned}$$

Thus

$$(1+a_0)^{-1} = 1 + u - 2u^2 + 10u^3 - 74u^4 + O(u^5),$$

where the $O(u^5)$ on $t \ge T_0$ is the Lagrange remainder of g. Both $a_0(u)$ and $a(t)$ lie in $[-1/2, 1/2]$, and so does the segment between them. On that segment the derivative of $(1+z)^{-1}$ has absolute value at most 4, so the mean value theorem gives

$$\left|(1+a)^{-1} - (1+a_0)^{-1}\right| \le 4|r(t)| \le 3780\, u^5.$$

Thus the same expansion holds for $(1+a)^{-1}$, with a possibly larger finite constant. Multiplying by t and subtracting t produces (A.5). No derivative of ρ_h is taken.

□

Lemma A.6. *There exist a finite $T_1 \ge T_0$ and finite positive constants C_σ, C_F, C_D such that for every $t \ge T_1$,*

$$\begin{aligned} \sigma(-t) &= t^{-2} - 6t^{-4} + 50t^{-6} + \rho_\sigma(t), & |\rho_\sigma| &\le C_\sigma t^{-8}, \\ F(-t) &= 2t^{-4} + \rho_F(t), & |\rho_F| &\le C_F t^{-6}, \\ D(-t) &= 4t^{-6} + \rho_D(t), & |\rho_D| &\le C_D t^{-8}. \end{aligned}$$

In particular there exists a finite T_* such that $F(-t) > 0$ and $D(-t) > 0$ for all $t \ge T_*$.

Proof. Retain $t \ge T_0$ and write $H = th(-t)$, so Lemma A.5 says

$$H = 1 - 2u + 10u^2 - 74u^3 + \tilde{\rho}, \qquad |\tilde{\rho}| \le C_h u^4.$$

Set $H_1 = H - 1 = -2u + 10u^2 - 74u^3 + \tilde{\rho}$. From $q = t + h = t(1 + uH)$ and $\sigma = 1 - qh$,

$$qh = H + uH^2, \qquad \sigma = 1 - H - uH^2 = -H_1 - u - 2uH_1 - uH_1^2.$$

Expand, using only products of series and the bound on $\tilde{\rho}$. The terms through u^3 are

$$\begin{aligned} -H_1 &= 2u - 10u^2 + 74u^3 + O(u^4), \\ -2uH_1 &= 4u^2 - 20u^3 + O(u^4), \\ H_1^2 &= 4u^2 - 40u^3 + O(u^4), \\ -uH_1^2 &= -4u^3 + O(u^4). \end{aligned}$$

Adding and using $-u$ gives

$$\sigma = u - 6u^2 + 50u^3 + O(u^4).$$

The summand $74u^3$ comes from the coefficient -74 in h. It is the reason that coefficient must be kept. The symbol $O(u^4)$ denotes a remainder bounded by a constant times u^4 for $t \geq T_0$; the constant exists because it is a finite combination of C_h and the absolute values of the displayed coefficients. This is the expansion of σ.

Next, $h^2 = uH^2 = u(1 + 2H_1 + H_1^2)$. Through order u^3,

$$h^2 = u - 4u^2 + 24u^3 + O(u^4).$$

The coefficient 24 uses h only through t^{-5}. Subtracting the expansion of σ,

$$F = h^2 - \sigma = 2u^2 + O(u^3) = 2t^{-4} + O(t^{-6}).$$

The leading coefficient 2 requires h through its $10t^{-5}$ term, which fixes the coefficient $-6t^{-4}$ in σ. Choose the constant C_F from this $O(t^{-6})$.

For D, square the expansion of σ:

$$\sigma^2 = u^2 - 12u^3 + O(u^4),$$

so $\sigma + \sigma^2 = u - 5u^2 + 38u^3 + O(u^4)$. Also

$$(1 - \sigma)h^2 = u - 5u^2 + 34u^3 + O(u^4),$$

where the coefficient 34 is $24 + 4 + 6$, coming from h^2, from $(-u)(-4u^2)$, and from $(6u^2)(u)$. Therefore

$$D = 4u^3 + O(u^4) = 4t^{-6} + O(t^{-8}).$$

The leading coefficient 4 uses the coefficient 50 in σ, hence uses the coefficient -74 in h. The constant C_D exists for the same reason as C_F.

Finally $F(-t) \geq t^{-4}(2 - C_F t^{-2})$ for $t \geq T_1$, and $D(-t) \geq t^{-6}(4 - C_D t^{-2})$ for $t \geq T_1$. Any

$$T_* > \max(T_1, \sqrt{C_F}, \sqrt{C_D/4})$$

is a finite cutoff on which both functions are positive. The value of T_* is not required.

□

A.4 First-zero argument

Lemma A.7. *Let $f \in C^1(\mathbb{R})$. Suppose there is a finite c_0 such that $f > 0$ on $(-\infty, c_0]$, and suppose $f'(c) > 0$ whenever $f(c) = 0$. Then $f > 0$ on $\mathbb{R}$.*

Proof. If the set $E = \{c : f(c) \leq 0\}$ is empty, there is nothing to prove. Otherwise $c_* = \inf E$ is finite and $c_* > c_0$. Continuity gives $f(c_*) = 0$, while $f > 0$ on $(-\infty, c_*)$. Differentiability gives

$$f(c_* - \varepsilon) = -\varepsilon f'(c_*) + o(\varepsilon).$$

The hypothesis $f'(c_*) > 0$ makes the right-hand side negative for small $\varepsilon > 0$, contradicting positivity on the left. The remainder in this one-variable Taylor expansion is the definition of $f'(c_*)$; it is not an asymptotic remainder of $M(t)$.

□

Theorem A.8. *The functions F and D are positive on all of $\mathbb{R}$. Consequently $\sigma' = qF > 0$ everywhere, so σ is strictly increasing, and $(h^2/\sigma)' = hD/\sigma^2 > 0$ everywhere, so h^2/σ is strictly increasing.*

Proof. Lemma A.6 supplies a finite left endpoint beyond which $F(-t) > 0$ and $D(-t) > 0$. On every zero of F, Subsection A.2 gives $F' = 2h\sigma > 0$. On every zero of D, Lemma A.3 gives $D' > 0$. Lemma A.7 applies to both functions. The sign statements for the derivatives of σ and h^2/σ were proved in Subsections A.1 and A.2, conditionally on $F > 0$ and $D > 0$.

Combined with Lemma A.2, the four scalar maps h, h^2, σ, and h^2/σ are strictly increasing on $\mathbb{R}$.

□

A.5 Differentiation under the integral and strict covariance

Let $w_{m,\mu}(c) = e^{\mu c}\phi(c)^2\Phi(c)^{m-2}$ and let $Z_m(\mu) = \int_{\mathbb{R}} w_{m,\mu}$. For each fixed m the map $\mu \mapsto w_{m,\mu}(c)$ is smooth. For $|\mu| \le M$, the estimate

$$|c|^N w_{m,\mu}(c) \le \frac{e^{M^2/2}}{2\pi}(1+|c|)^N e^{-c^2/2}$$

follows from $M|c| \le (c^2 + M^2)/2$. It gives a common integrable bound for every fixed polynomial degree.

Lemma A.9. *For every integer $m \ge 3$ and every real μ, the normalizing constant Z_m is smooth and positive, and if $f : \mathbb{R} \to \mathbb{R}$ is continuous and $|f(c)| \le C(1+|c|)^N$, then $\mu \mapsto \mathbb{E}_{m,\mu}[f]$ is smooth and*

$$\frac{d}{d\mu}\mathbb{E}_{m,\mu}[f(c)] = \mathrm{Cov}_{m,\mu}(f(c), c).$$

If in addition f is strictly increasing, the covariance is strictly positive.

Proof. The difference quotient of $w_{m,\mu}$ in μ is $w_{m,\mu}(c) \cdot c \cdot e^{\theta c}/e^{\mu c}$ for an intermediate θ, hence is dominated on a neighborhood of any fixed μ by an integrable function of the type above. Differentiation under the integral sign is therefore legitimate and gives $Z_m'(\mu) = \int c\, w_{m,\mu}$. The same domination applies to $f(c)w_{m,\mu}(c)$ and to $cf(c)w_{m,\mu}(c)$, so the quotient defining the expectation may be differentiated by the quotient rule. The result is the covariance.

For the strict sign, let X, Y be independent copies of the law $p_{m,\mu}$. Then

$$\mathrm{Cov}(f(X), X) = \frac{1}{2}\mathbb{E}[(f(X) - f(Y))(X - Y)].$$

The integrand is nonnegative. It is not almost surely zero: the density $p_{m,\mu}$ is continuous and strictly positive on $\mathbb{R}$, so $P(X \ne Y) = 1$, and a strictly increasing f then gives a strictly positive integrand on $\{X \ne Y\}$. The expectations exist by the polynomial-growth hypothesis and the Gaussian tail of $p_{m,\mu}$.

□

Lemma A.10. *For each integer $m \ge 3$, the maps a_m, b_m, H_m, and a_m/b_m are smooth and strictly increasing on $\mathbb{R}$. In particular they are strictly increasing on $[0, \infty)$, and $H_m(\mu) \ge H_m(0) > 0$ for $\mu \ge 0$.*

Proof. The functions h^2, σ, and h are continuous, strictly increasing, and of polynomial growth by Lemma A.2 and Theorem A.8. Lemma A.9 gives the result for a_m, b_m, and H_m. Strict increase of H_m yields $H_m(\mu) \ge H_m(0)$. Positivity of h yields $H_m(0) > 0$.

For the ratio, let $Q_{m,\mu}$ be the probability with density proportional to $\sigma(c)w_{m,\mu}(c)$. The factor $\sigma(c)$ does not depend on μ, lies in $(0, 1)$, and is continuous and positive, so $Q_{m,\mu}$ is again a smooth exponential family in μ with the same dominating Gaussian tails. Moreover

$$\mathbb{E}_{Q_{m,\mu}}\Big[\frac{h^2}{\sigma}\Big] = \frac{\mathbb{E}_{m,\mu}[h^2]}{\mathbb{E}_{m,\mu}[\sigma]} = \frac{a_m(\mu)}{b_m(\mu)}.$$

Theorem A.8 says that h^2/σ is strictly increasing. Lemma A.6 gives $h(-t) \sim t^{-1}$ and $\sigma(-t) \sim t^{-2}$, so $h^2/\sigma \to 1$ as $c \to -\infty$. For $c \geq 0$, the same theorem gives $\sigma(c) \geq \sigma(0) = 1 - 2/\pi > 0$, and Lemma A.2 gives $h(c) \leq c + \sqrt{2/\pi}$. Hence

$$0 < \frac{h(c)^2}{\sigma(c)} \leq \frac{(c + \sqrt{2/\pi})^2}{1 - 2/\pi}$$

on $[0, \infty)$. Continuity controls the remaining compact interval, so the ratio has polynomial growth on all of $\mathbb{R}$. The weight of $Q_{m,\mu}$ is $\sigma(c) w_{m,\mu}(c)$. Since $0 < \sigma < 1$, every integrable dominant used for $w_{m,\mu}$ dominates the corresponding integrand for $Q_{m,\mu}$, and differentiation in μ still multiplies that weight by c. The covariance argument of Lemma A.9 therefore applies to $Q_{m,\mu}$ and gives strict increase in μ.

□

Lemma A.11 (monotonicity in the number of signals)**.** *Fix $\mu \in \mathbb{R}$. For every integer $m \geq 3$,*

$$b_{m+1}(\mu) > b_m(\mu).$$

The same strict increase holds for $a_m(\mu)$, $H_m(\mu)$, and $a_m(\mu)/b_m(\mu)$.

Proof. The un-normalized weight at $m+1$ is the weight at m multiplied by $\Phi(c)$. The function Φ is strictly increasing, and σ is strictly increasing. For a positive probability P with full support on $\mathbb{R}$ and a bounded, positive, strictly increasing g, the reweighted expectation of a strictly increasing integrable f satisfies

$$\frac{\mathbb{E}[fg]}{\mathbb{E}[g]} - \mathbb{E}[f] = \frac{1}{2\mathbb{E}[g]} \mathbb{E}\big[(f(X) - f(Y))(g(X) - g(Y))\big] > 0,$$

by the same independent-copy argument as in Lemma A.9. Apply this to $P = p_{m,\mu}$, $g = \Phi$, and $f = \sigma$. Taking $f = h^2$ or $f = h$ proves the corresponding assertions for a_m and H_m. The ratio a/b is the expectation of the strictly increasing function h^2/σ under the σ-reweighted law, and increasing m multiplies that law by another strictly increasing copy of Φ. The polynomial-growth integrability is the same Gaussian estimate as before.

□

A.6 The four-signal benchmark

Write $\mathcal{I}_4 = \int_{\mathbb{R}} \phi^4$ and $\mathcal{J} = \int_{\mathbb{R}} \phi^2 \Phi^2$.

Lemma A.12. *$\mathcal{I}_4 = 1/(4\sqrt{2}\,\pi^{3/2})$ and*

$$b_4(0) = 1 - \frac{4\mathcal{I}_4}{3\mathcal{J}}.$$

Proof. The identity $\phi(c)^4 = (2\pi)^{-2} e^{-2c^2}$ and $\int e^{-2c^2}\, dc = \sqrt{\pi/2}$ give

$$\mathcal{I}_4 = (2\pi)^{-2} \sqrt{\pi/2} = 1/(4\sqrt{2}\,\pi^{3/2}).$$

For the expectation, $\sigma = 1 - qh$ and $qh\,\phi^2\Phi^2 = c\phi^3\Phi + \phi^4$. Integrate $c\phi^3\Phi$ by parts. From $(\phi^3)' = -3c\phi^3$ and the vanishing of $\phi^3\Phi$ at both ends,

$$\int c\phi^3\Phi = \frac{1}{3} \int \phi^4.$$

Hence $\int qh\,\phi^2\Phi^2 = (4/3)\mathcal{I}_4$, and dividing by $\mathcal{J}$ produces the formula. This identity is exact; the arcsin evaluation of $\mathcal{J}$ is not used.

□

Lemma A.13. $\mathcal{J} \geq (3\pi+2)/(24\pi^{3/2})$, *and therefore*

$$b_4(0) \geq 1 - \frac{4\sqrt{2}}{3\pi+2}.$$

Proof. Expand $\Phi^2 = (\Phi - 1/2)^2 + (\Phi - 1/2) + 1/4$. The cross term $\phi^2(\Phi - 1/2)$ is an odd integrable function, so its integral vanishes. Also $\int \phi^2 = 1/(2\sqrt{\pi})$, hence

$$\mathcal{J} = \frac{1}{4}\int \phi^2 + \int \phi^2(\Phi - 1/2)^2 = \frac{1}{8\sqrt{\pi}} + \int \phi^2(\Phi - 1/2)^2.$$

Integration by parts, using $(\phi^2)' = -2c\phi^2$ and vanishing boundary terms, gives

$$\int c\phi^2(\Phi - 1/2) = \frac{1}{2}\int \phi^3.$$

Cauchy–Schwarz in the form

$$\Big(\int (c\phi)\cdot(\phi(\Phi - 1/2))\Big)^2 \leq \Big(\int c^2\phi^2\Big)\Big(\int \phi^2(\Phi - 1/2)^2\Big)$$

therefore yields a lower bound on the second integral. The Gaussian integrals

$$\int \phi^3 = \frac{1}{2\pi\sqrt{3}}, \qquad \int c^2\phi^2 = \frac{1}{4\sqrt{\pi}}$$

are elementary. The resulting quotient is $1/(12\pi^{3/2})$, and

$$\mathcal{J} \geq \frac{1}{8\sqrt{\pi}} + \frac{1}{12\pi^{3/2}} = \frac{3\pi+2}{24\pi^{3/2}}.$$

Cauchy–Schwarz is not asserted to be equality. Because $\mathcal{I}_4 > 0$, replacing $\mathcal{J}$ by this lower bound increases $4\mathcal{I}_4/(3\mathcal{J})$. The arithmetic

$$\frac{4\mathcal{I}_4}{3} \Big/ \frac{3\pi+2}{24\pi^{3/2}} = \frac{4\sqrt{2}}{3\pi+2}$$

is the comparison of Lemma A.12 with Lemma A.13's bound.

□

Lemma A.14. $\pi > 28/9$ *and* $\sqrt{2} < 17/12$. *Consequently* $4\sqrt{2}/(3\pi+2) < 1/2$, *and* $b_4(0) > 1/2$.

Proof. Let $\theta = \arctan(1/5)$. Then $0 < \theta < \pi/4$, because $0 < 1/5 < 1$ and tangent is increasing on $(0, \pi/2)$. The double-angle formula applies and gives

$$\tan(2\theta) = \frac{2\cdot(1/5)}{1-(1/5)^2} = \frac{5}{12}.$$

Thus $2\theta \in (0, \pi/2)$. Since $\tan(2\theta) < 1$, one has $2\theta \in (0, \pi/4)$, and the formula applies again:

$$\tan(4\theta) = \frac{2\cdot(5/12)}{1-(5/12)^2} = \frac{120}{119},$$

with $4\theta \in (0, \pi/2)$. Set $\eta = 4\theta$ and $\beta = \arctan(1/239) \in (0, \pi/4)$. Then $\tan\eta = 120/119 > 1/239 = \tan\beta$. Both angles lie in $(0, \pi/2)$ and tangent is increasing there, so $\eta > \beta$ and $\eta - \beta \in (0, \pi/2)$. Therefore

$$\begin{aligned}\tan(\eta-\beta) &= \frac{120/119 - 1/239}{1 + (120/119)(1/239)} = \frac{120\cdot 239 - 119}{119\cdot 239 + 120} \\ &= \frac{28680 - 119}{28441 + 120} = \frac{28561}{28561} = 1.\end{aligned}$$

The unique angle in $(0, \pi/2)$ with tangent 1 is $\pi/4$, so

$$\frac{\pi}{4} = 4 \arctan \frac{1}{5} - \arctan \frac{1}{239}.$$

For every real t, the identity $(1 - t^2)(1 + t^2) + t^4 = 1$ rearranges to

$$\frac{1}{1 + t^2} = 1 - t^2 + \frac{t^4}{1 + t^2}.$$

Integrating from 0 to $x > 0$ gives

$$\arctan x = x - \frac{x^3}{3} + \int_0^x \frac{t^4}{1 + t^2}\, dt > x - \frac{x^3}{3},$$

the integrand being positive. In particular

$$\arctan \frac{1}{5} > \frac{1}{5} - \frac{1}{3 \cdot 5^3} = \frac{74}{375}.$$

The same representation gives $\arctan y < y$ for $y > 0$, because $1/(1 + t^2) < 1$. Thus $\arctan(1/239) < 1/239$. Combining these strict bounds with Machin's formula,

$$\frac{\pi}{4} > 4 \cdot \frac{74}{375} - \frac{1}{239} = \frac{296}{375} - \frac{1}{239} = \frac{70369}{89625}.$$

The cross multiplication $70369 \cdot 9 - 7 \cdot 89625 = 5946 > 0$ gives $70369/89625 > 7/9$, so $\pi > 28/9$. Both arctangent inequalities are strict.

Separately, $(17/12)^2 = 289/144 > 288/144 = 2$, so $\sqrt{2} < 17/12$. Hence

$$8\sqrt{2} < 8 \cdot \frac{17}{12} = \frac{34}{3}, \qquad 3\pi + 2 > 3 \cdot \frac{28}{9} + 2 = \frac{34}{3}.$$

The second inequality is strict, so $8\sqrt{2} < 3\pi + 2$ and $4\sqrt{2}/(3\pi + 2) < 1/2$. Lemma A.13 gives $b_4(0) \geq 1 - 4\sqrt{2}/(3\pi + 2) > 1/2$.

□

Corollary A.15. *For every integer $m \geq 4$ and every $\mu \geq 0$,*

$$\frac{1}{2} < b_m(\mu) < 1.$$

Proof. The upper bound holds for every m and μ because $0 < \sigma < 1$ pointwise and $p_{m,\mu}$ has full support, so the expectation of σ cannot attain 1. The lower bound holds at $(m, \mu) = (4, 0)$ by Lemma A.14. Lemmas A.10 and A.11 move the value strictly upward in each variable separately, and the inequality is already strict at the minimal corner.

□

A.7 The balance identity and the initial values

Lemma A.16. *For every integer $m \geq 3$ and every real μ,*

$$2a_m(\mu) + (m - 1)b_m(\mu) = m + \mu H_m(\mu). \tag{A.6}$$

Proof. The logarithmic derivative of the un-normalized weight is

$$\mu - 2c + (m - 2)q = \mu - 2h + mq,$$

which is the weight's formula $p'/p = \mu - 2h + mq$. Fix $R > 0$. Then

$$\int_{-R}^{R} (hp_{m,\mu})' = [hp_{m,\mu}]_{-R}^{R}.$$

The bound of Lemma A.2 and the Gaussian tail of $p_{m,\mu}$ force the boundary term to 0 as $R \to \infty$. Expanding the derivative,

$$(hp)' = \sigma p + h(\mu - 2h + mq)p.$$

The resulting integrand is absolutely dominated by an integrable multiple of $(1 + c^2)p_{m,\mu}(c)$, because $0 < \sigma < 1$, $qh = 1 - \sigma$, and h has linear growth. Passing to the limit,

$$b + \mu H - 2a + m\mathbb{E}[qh] = 0.$$

The identity $\mathbb{E}[qh] = 1 - b$ rearranges into (A.6).

□

Lemma A.17. *Fix an integer $m \geq 4$. At $\mu = 0$,*

$$a(0) < \frac{m}{2}, \qquad \frac{a(0)}{b(0)} < m, \qquad a(0) + (m-2)b(0) < m - 1.$$

All three inequalities are strict.

Proof. The identity at $\mu = 0$ is $2a + (m-1)b = m$, so $a = [m - (m-1)b]/2$. Since $b > 0$, one has $a < m/2$. Next

$$\frac{a}{b} = \frac{m}{2b} - \frac{m-1}{2}.$$

The inequality $a/b < m$ rearranges to $b > m/(3m-1)$. For $m > 1$, $m/(3m-1) < 1/2$, and Corollary A.15 supplies $b > 1/2$. Finally

$$a + (m-2)b = \frac{m + (m-3)b}{2}.$$

Comparing with $m - 1$ rearranges to $(m-3)b < m - 2$. For $m = 4$ the right side is 2 and the left side is $b < 1$. For $m > 4$, $(m-2)/(m-3) = 1 + 1/(m-3) > 1 > b$. Every comparison is strict.

□

A.8 Growth and the ordered crossings

Lemma A.18. *For each integer $m \geq 3$ and every $\mu \geq 0$,*

$$2a_m(\mu) > 1 + \mu H_m(0).$$

In particular $a_m(\mu) \to +\infty$ as $\mu \to +\infty$. For $m = 3$, $\mu H_3(\mu) \geq \mu H_3(0)$, so $\mu H_3(\mu) \to +\infty$.

Proof. By (A.6) and $b_m(\mu) < 1$,

$$\begin{aligned} 2a_m(\mu) &= m + \mu H_m(\mu) - (m-1)b_m(\mu) \\ &> m + \mu H_m(\mu) - (m-1) = 1 + \mu H_m(\mu). \end{aligned}$$

Lemma A.10 gives $H_m(\mu) \geq H_m(0) > 0$, and the displayed bound follows. The right-hand side tends to infinity. For $m = 3$ the same lemma gives the linear lower bound $\mu H_3(0)$.

□

Lemma A.19 (comparison of the three coefficients)**.** *Let $m \geq 4$, $a > 0$, and $1/2 < b < 1$. Then*

$$a = \frac{m}{2} \Longrightarrow L_0(m,a) = L_\perp(m) \text{ and } L_\perp(m) < L_1(m,a,b),$$

$$a = mb \Longrightarrow L_0(m,a) = L_1(m,a,b) \text{ and } L_\perp(m) < L_0(m,a),$$

$$\min\big(L_0(m,a), L_\perp(m)\big) < L_1(m,a,b).$$

Proof. If $a = m/2$, then $L_0 = (m-1)/2 = L_\perp$ and

$$L_1 - L_\perp = \frac{m-2}{2}\Big(b - \frac{1}{2}\Big) > 0.$$

If $a = mb$, then $L_1 = (mb + (m-2)b)/2 = (m-1)b = L_0$, while

$$L_\perp - L_0 = (m-1)\Big(\frac{1}{2} - b\Big) < 0.$$

For the third claim, the same algebra gives the unrestricted identity

$$L_1 - L_0 = \frac{m-2}{2m}(mb - a) \qquad (m > 2).$$

Suppose toward a contradiction that $L_1 \leq L_0$ and $L_1 \leq L_\perp$. The first forces $a \geq mb$. The second forces $a + (m-2)b \leq m - 1$. Adding these,

$$a + (m-2)b \geq mb + (m-2)b = 2(m-1)b > (m-1),$$

which contradicts the second inequality. The last step is strict because $b > 1/2$.

□

Theorem A.20 (unique positive crossings for $m \geq 4$)**.** *Fix an integer $m \geq 4$. There exist unique real numbers*

$$0 < \mu_{1\perp}(m) < \mu_{0\perp}(m) < \mu_{01}(m) < \infty$$

such that

$$\begin{aligned} a(\mu_{1\perp}) + (m-2)b(\mu_{1\perp}) &= m - 1, \\ a(\mu_{0\perp}) &= m/2, \\ a(\mu_{01}) &= m\, b(\mu_{01}), \end{aligned}$$

and each equation has no other root in $[0,\infty)$. Equivalently, the three normalized curvatures satisfy

$$L_1 = L_\perp, \qquad L_0 = L_\perp, \qquad L_0 = L_1$$

at these three parameters, in that order.

Proof. The maps

$$f_{1\perp} = a + (m-2)b, \qquad f_{0\perp} = a, \qquad f_{01} = a/b$$

are continuous and strictly increasing on $[0,\infty)$ by Lemma A.10. Lemma A.17 says that their values at 0 are strictly less than $m-1$, $m/2$, and m respectively. Lemma A.18 and the bound $1/2 < b < 1$ give

$$f_{0\perp}(\mu) \to \infty, \qquad f_{01}(\mu) \geq a(\mu) \to \infty, \qquad f_{1\perp}(\mu) \geq a(\mu) \to \infty.$$

The intermediate-value theorem supplies at least one positive root of each equation, and strict increase supplies uniqueness.

Let $\mu_{0\perp}$ be the unique root of $a = m/2$. Corollary A.15 gives $b(\mu_{0\perp}) > 1/2$, so $a(\mu_{0\perp}) < mb(\mu_{0\perp})$. Thus $f_{01}(\mu_{0\perp}) < m$, and the unique root μ_{01} of $f_{01} = m$ lies strictly to the right of $\mu_{0\perp}$. At the same point,

$$a + (m-2)b = \frac{m}{2} + (m-2)b > \frac{m}{2} + \frac{m-2}{2} = m - 1,$$

so $f_{1\perp}(\mu_{0\perp}) > m - 1$. The unique root $\mu_{1\perp}$ lies strictly to the left of $\mu_{0\perp}$.

□

A.9 Ordering of the decoding curvatures

For $\lambda > 0$ set $\mu = \lambda/\sqrt{\alpha}$ and

$$\omega_\lambda(c) = \frac{\mu e^{-\mu^2/2}}{m}\, e^{\mu c}\phi(c)^2\Phi(c)^{m-2}.$$

Let $A_\lambda = \int \omega_\lambda h^2$, $B_\lambda = \int \omega_\lambda \sigma$, and $C_\lambda = \int \omega_\lambda$. The change of scale between $\omega_\lambda\, dc$ and $p_{m,\mu}\, dc$ is a positive constant, so $C_\lambda > 0$ and

$$\frac{A_\lambda}{C_\lambda} = a(\mu), \qquad \frac{B_\lambda}{C_\lambda} = b(\mu).$$

Define

$$\Lambda_0 = \alpha A_\lambda, \qquad \Lambda_1 = \frac{A_\lambda + (m-2)B_\lambda}{2}, \qquad \Lambda_\perp = \frac{m-1}{2}C_\lambda.$$

Dividing by C_λ produces exactly L_0, L_1, and $L_\perp$. Equality and strict inequality of the Λ's are therefore the same as those of the L's.

Theorem A.21 (ordering)**.** *Fix an integer $m \geq 4$ and $\lambda > 0$, and let $\mu = \lambda/\sqrt{\alpha}$. On the four open intervals determined by Theorem A.20,*

$$\begin{aligned} 0 < \mu < \mu_{1\perp} &\Longrightarrow \Lambda_0 < \Lambda_1 < \Lambda_\perp, \\ \mu_{1\perp} < \mu < \mu_{0\perp} &\Longrightarrow \Lambda_0 < \Lambda_\perp < \Lambda_1, \\ \mu_{0\perp} < \mu < \mu_{01} &\Longrightarrow \Lambda_\perp < \Lambda_0 < \Lambda_1, \\ \mu > \mu_{01} &\Longrightarrow \Lambda_\perp < \Lambda_1 < \Lambda_0. \end{aligned}$$

At $\mu_{1\perp}$ one has $\Lambda_1 = \Lambda_\perp$ and Λ_0 strictly smaller than both. At $\mu_{0\perp}$ one has $\Lambda_0 = \Lambda_\perp$ and Λ_1 strictly larger than both. At μ_{01} one has $\Lambda_0 = \Lambda_1$ and $\Lambda_\perp$ strictly smaller than both. The corresponding thresholds in the variable λ are $\sqrt{\alpha}$ times the thresholds in μ.

Proof. Strict increase of $a + (m-2)b$, of a, and of a/b converts the three roots into sign changes of $\Lambda_1 - \Lambda_\perp$, of $\Lambda_0 - \Lambda_\perp$, and of $\Lambda_0 - \Lambda_1$, respectively. The order of the roots forbids every other permutation. Lemma A.19 is the endpoint comparison: when $a = m/2$, $b > 1/2$ forces $\Lambda_\perp < \Lambda_1$, and when $a = mb$, the same bound forces $\Lambda_\perp < \Lambda_0$. The relation $\lambda = \mu\sqrt{\alpha}$ is the definition $\mu = \lambda/\sqrt{\alpha}$.

□

Corollary A.22 (the strict normal region)**.** *For $m \geq 4$ and $\lambda > 0$, the normal curvature is strictly smaller than both in-plane curvatures if and only if $\lambda/\sqrt{\alpha} > \mu_{0\perp}(m)$. Moreover,*

$$\min(\Lambda_0, \Lambda_\perp) < \Lambda_1$$

for every positive channel parameter.

Proof. The equivalence follows from Theorem A.21, including its endpoint comparisons. The strict inequality is Lemma A.19 multiplied by $C_\lambda > 0$. □

A.10 The triangular case

For $m = 3$ the summand $\mathcal{H}_0$ is absent. The identity (A.6) becomes $2a + 2b = 3 + \mu H_3(\mu)$, so

$$\frac{\Lambda_1}{C_\lambda} = \frac{a+b}{2} = \frac{3+\mu H_3(\mu)}{4}, \qquad \frac{\Lambda_\perp}{C_\lambda} = 1.$$

Thus $\Lambda_1 < \Lambda_\perp$, $\Lambda_1 = \Lambda_\perp$, or $\Lambda_1 > \Lambda_\perp$ according as $\mu H_3(\mu) < 1$, $= 1$, or > 1.

Theorem A.23. *The function $\mu \mapsto \mu H_3(\mu)$ is continuous and strictly increasing from $[0, \infty)$ onto $[0, \infty)$. It has a unique positive root $\mu_*(3)$ of the equation $\mu H_3(\mu) = 1$. For $0 < \mu < \mu_*$ the in-plane coefficient is strictly smaller; for $\mu > \mu_*$ the normal coefficient is strictly smaller; at μ_* they agree.*

Proof. Lemma A.9 gives $H_3'(\mu) = \mathrm{Cov}(h, c) > 0$. Hence

$$\frac{d}{d\mu}(\mu H_3(\mu)) = H_3(\mu) + \mu H_3'(\mu) > 0$$

for every $\mu \geq 0$, the value $H_3(\mu)$ being positive. The function vanishes at 0. Lemma A.18 sends it to infinity. The intermediate-value theorem and strict increase give a unique positive root, and the sign chart is the sentence above.

□

B Proof of the exact normal-shell theorem

We prove Theorems 15 and 16. Throughout, $m \geq 3$, $n = m - 1$, $\alpha = n/m$, $J = \mathbf{1}\mathbf{1}^\mathsf{T}$, $P = I_m - J/m$, and $G_\Delta = P/\alpha$. The reference configuration $V = [v_1 \ \cdots \ v_m]$ satisfies $V^\mathsf{T} V = G_\Delta$, and $d \geq m$. Fix $\lambda > 0$ and put $\mu = \lambda/\sqrt{\alpha}$. The scalar results in Appendix A give the strict normal regime: $a_m(\mu) > m/2$ if $m \geq 4$, and $a_3(\mu) + b_3(\mu) > 2$ if $m = 3$.

We use coordinates on the positive square root of the centered Gram matrix. They give a polynomial parametrization of the extended shell objective; no differentiable choice of eigenvectors or implicit shell equation is needed. The objective is extended to a full neighborhood in a fixed finite-dimensional matrix space before it is restricted to realizable configurations.

B.1 A smooth extension of the actual decoding deficit

For a centered Gaussian vector ξ with covariance G, the success probability of a unit configuration is

$$P_{\mathrm{c}}(G; \lambda) = \frac{e^{-\lambda^2/2}}{m} \mathbb{E} e^{\lambda \max_i \xi_i}. \tag{B.1}$$

This formula is valid when G is singular. If Z is a standard Gaussian in the signal space, the score vector $(\langle x_i, Z \rangle)_{i=1}^m$ has exactly covariance G.

Let b be an independent standard real Gaussian. The vector $\sqrt{\alpha}\,\xi + m^{-1/2} b \mathbf{1}$ has covariance $R = \alpha G + J/m$. Since $\mu\sqrt{\alpha} = \lambda$,

$$\mathbb{E}_R e^{\mu \max_i X_i} = e^{\mu^2/(2m)} \mathbb{E}_G e^{\lambda \max_i \xi_i}, \qquad e^{-\mu^2/2} e^{\mu^2/(2m)} = e^{-\lambda^2/2}.$$

Consequently the actual deficit is the restriction of

$$\mathcal{F}_\lambda(R) = \frac{e^{-\mu^2/2}}{m} \left(\mathbb{E}_{I_m} e^{\mu \max_i Z_i} - \mathbb{E}_R e^{\mu \max_i X_i} \right) : \qquad \mathcal{F}_\lambda(\alpha G + J/m) = P_{\mathrm{c}}(G_\Delta; \lambda) - P_{\mathrm{c}}(G; \lambda). \tag{B.2}$$

The expectation defining this extension makes sense for every positive definite symmetric R near I_m, including matrices for which $(R - J/m)/\alpha$ is not positive semidefinite.

Here is the regularity justification used below. On a closed covariance neighborhood satisfying

$$\tfrac{1}{2} I_m \preceq R \preceq 2 I_m, \tag{B.3}$$

write

$$p_R(x) = (2\pi)^{-m/2} (\det R)^{-1/2} \exp(-x^{\mathsf{T}} R^{-1} x/2).$$

A derivative of order k in the covariance entries is $P_{k,R}(x) p_R(x)$, where $P_{k,R}$ is a polynomial of degree at most $2k$ whose coefficients are uniformly bounded on this neighborhood. Hence

$$|\partial_R^\beta p_R(x)| \le K_\beta (1 + \|x\|^{2|\beta|}) e^{-\|x\|^2/4}. \tag{B.4}$$

Multiplication by $e^{\mu \max_i x_i} \le e^{\mu \|x\|}$ preserves integrability. Dominated differentiation therefore proves that $\mathcal{F}_\lambda$ is smooth on a smaller neighborhood. In particular it is C^2 jointly in all its covariance parameters, and its third derivatives are locally bounded. All these neighborhoods and bounds depend only on m and λ.

For clarity, both bounds in (B.3) are used: the upper bound supplies the uniform Gaussian decay in (B.4). The argument estimates derivatives of the density, rather than asserting a polynomial bound, uniform in x, on a density ratio.

B.2 The first two covariance variations

Let $A = A^{\mathsf{T}}$ have zero diagonal, and set

$$S(A) = \sum_{i<j} A_{ij} = \tfrac{1}{2} \mathbf{1}^{\mathsf{T}} A \mathbf{1}, \qquad N(A) = \|A\|_F^2, \qquad R_1(A) = \|A\mathbf{1}\|^2.$$

We recall the scalar notation

$$q(c) = \frac{\phi(c)}{\Phi(c)}, \qquad h(c) = c + q(c), \qquad \sigma(c) = 1 - q(c)h(c), \qquad k(c) = \phi(c)^2 \Phi(c)^{m-2},$$

and define the positive weight and its moments by

$$\begin{aligned} \omega_\lambda(c) &= \frac{\mu e^{-\mu^2/2}}{m} e^{\mu c} k(c), & C_\lambda &= \int_{\mathbb{R}} \omega_\lambda(c)\, dc, \\ A_\lambda &= \int_{\mathbb{R}} \omega_\lambda(c) h(c)^2\, dc, & B_\lambda &= \int_{\mathbb{R}} \omega_\lambda(c) \sigma(c)\, dc. \end{aligned} \tag{B.5}$$

Thus $C_\lambda > 0$, $A_\lambda / C_\lambda = a_m(\mu)$, and $B_\lambda / C_\lambda = b_m(\mu)$.

Lemma B.1 (Full covariance expansion)**.** *For symmetric zero-diagonal A near zero,*

$$\begin{aligned} \mathcal{F}_\lambda(I_m + A) = C_\lambda S(A) + \frac{A_\lambda}{4} N(A) - \frac{C_\lambda - B_\lambda}{2} R_1(A) \\ + \frac{S(A)^2}{2} \int_{\mathbb{R}} \omega_\lambda(c) q(c)^2\, dc + O(\|A\|_F^3). \end{aligned} \tag{B.6}$$

The remainder is uniform on a sufficiently small matrix neighborhood.

Proof. At the identity, the density Taylor coefficients are

$$p_{I_m + A}(x) = p_{I_m}(x) [1 + \tfrac{1}{2} x^{\mathsf{T}} A x + \tfrac{1}{4} N(A) + \tfrac{1}{8} (x^{\mathsf{T}} A x)^2 - \tfrac{1}{2} x^{\mathsf{T}} A^2 x] + \mathcal{R}_A(x). \tag{B.7}$$

This follows by expanding $(I_m + A)^{-1}$ and $-\frac{1}{2}\log\det(I_m + A)$, using $\operatorname{tr} A = 0$. The density differentiation argument above gives, after shrinking the neighborhood,

$$|\mathcal{R}_A(x)| \le K\,\|A\|_F^3\,(1 + \|x\|^6)e^{-\|x\|^2/4}.$$

It thus also gives the integrated third-order remainder.

We identify the coefficients by integrating over $(-\infty, c]^m$. The three one-dimensional truncated moments needed are

$$\int_{-\infty}^{c} \phi = \Phi(c), \qquad \int_{-\infty}^{c} x\phi(x)\,dx = -\phi(c), \qquad \int_{-\infty}^{c} x^2\phi(x)\,dx = \Phi(c) - c\phi(c).$$

The linear coefficient of A_{ij}, $i < j$, is $k(c)$. For the quadratic coefficients, an edge square A_{ij}^2 contributes $c^2k(c)/2$; a product of distinct edges with one common vertex contributes $-cq(c)k(c)$; and a product of disjoint edges contributes $q(c)^2k(c)$. These assertions follow directly from (B.7): the corresponding polynomials are

$$\tfrac{1}{2}(1 - x_i^2)(1 - x_j^2), \qquad (x_i^2 - 1)x_jx_\ell, \qquad x_ix_jx_\ell x_s,$$

respectively. Each product of distinct edges is counted once.

Put $Q_e = \sum_{i<j} A_{ij}^2 = N(A)/2$, and denote the sums of adjacent and disjoint edge products by U_e and W_e. Then

$$R_1(A) = 2Q_e + 2U_e, \qquad S(A)^2 = Q_e + 2U_e + 2W_e.$$

Substitution gives the fixed-c expansion

$$\frac{F_{I_m+A}(c) - \Phi(c)^m}{k(c)} = S(A) + \frac{h(c)^2}{4}N(A) - \frac{q(c)h(c)}{2}R_1(A) + \frac{q(c)^2}{2}S(A)^2 + O_c(\|A\|_F^3). \tag{B.8}$$

We justify the subsequent integration, including its sign. Expressing the expectations through the distribution of the maximum gives

$$\mathbb{E}_{I_m} e^{\mu \max_i Z_i} - \mathbb{E}_R e^{\mu \max_i X_i} = \mu \int_{\mathbb{R}} e^{\mu c}\big(F_R(c) - \Phi(c)^m\big)\,dc. \tag{B.9}$$

One may obtain it by integration by parts, or by applying Fubini to the difference of the two laws, which has total mass zero. The same argument applies to a positive-order covariance derivative of the density: its integral over all of $\mathbb{R}^m$ is zero. For $c \ge 0$, its integral over $(-\infty, c]^m$ is minus its integral over the complementary set, which is contained in $\{\|x\| \ge c\}$. For $c \le 0$, the orthant itself lies in $\{\|x\| \ge |c|\}$. Formula (B.4) therefore bounds each first or second orthant derivative by $K(1 + |c|^N)e^{-c^2/8}$. This remains integrable after multiplication by $e^{\mu c}$. It validates Fubini and differentiation of (B.9) through order two.

Multiply the explicitly displayed normalized coefficients in (B.8) by the full weight $\omega_\lambda(c)$, which includes $k(c)$, and integrate. Since $qh = 1 - \sigma$, this yields (B.6). The bound $q(c) \le |c| + \sqrt{2/\pi}$ ensures finiteness of the q^2 coefficient. The integrated remainder comes from the smooth density argument, not from integrating an unspecified fixed-c remainder. □

B.3 Compression and its exact inverse

Identify operators on $H = \mathbf{1}^\perp$ with symmetric $m \times m$ matrices C satisfying $PC = CP = C$. Define the linear map

$$\mathcal{U}(C)_{ij} = C_{ij} - \tfrac{1}{2}(C_{ii} + C_{jj}). \tag{B.10}$$

It has zero diagonal, and direct multiplication gives

$$P\mathcal{U}(C)P = C, \qquad \mathcal{U}(PAP) = A \quad \text{if } A = A^{\mathsf{T}},\ \operatorname{diag} A = 0. \tag{B.11}$$

For the second identity, write $a = A\mathbf{1}$ and $s = \mathbf{1}^\mathsf{T} A\mathbf{1}$. The entries of PAP are $A_{ij} - (a_i + a_j)/m + s/m^2$, and its diagonal entries are $-2a_i/m + s/m^2$; (B.10) cancels the extra terms. Since a centered matrix has zero total entry sum,

$$S(\mathcal{U}(C)) = -\frac{m}{2}\operatorname{tr} C. \tag{B.12}$$

The shape space used in this appendix is

$$\mathcal{D}_m = \{D = D^\mathsf{T} : PD = DP = D,\ \operatorname{tr} D = 0\}.$$

If $D \in \mathcal{D}_m$ and $B = \mathcal{U}(D)$, then $S(B) = 0$ and $D = PBP$. Writing $b = B\mathbf{1}$, we have $\mathbf{1}^\mathsf{T} b = 0$ and

$$D = B - \frac{1}{m}(b\mathbf{1}^\mathsf{T} + \mathbf{1}b^\mathsf{T}).$$

Expansion of the squared Frobenius norm gives the useful exact identity

$$\operatorname{tr}(D^2) = \|D\|_F^2 = N(B) - \frac{2}{m}R_1(B). \tag{B.13}$$

B.4 The actual Procrustes infimum and a uniform neighborhood

For unit configurations define

$$d_P(X, V)^2 = \inf_{U \in O(d)} \|X - UV\|_F^2 .$$

This is the actual configuration distance. The following calculation both evaluates the infimum near a simplex and exhibits an attaining orthogonal map.

Let $G = X^\mathsf{T} X$, and set

$$T = (\alpha PGP)^{1/2}, \qquad K = T + (I_m - P). \tag{B.14}$$

The square root is positive semidefinite, $PT = TP = T$, and $T^2 = \alpha PGP$. Assume for the moment that K is invertible. Put

$$W = XPK^{-1}.$$

The regular Gram identity implies $V\mathbf{1} = 0$. The matrix K preserves H and its orthogonal complement; on these spaces it acts as T and the identity, respectively. Consequently

$$W^\mathsf{T} W = K^{-1}PGPK^{-1} = \alpha^{-1}P = G_\Delta, \qquad WP = W, \qquad XP = WT.$$

Configurations with the same Gram matrix are related by an orthogonal map of the ambient space. Indeed, sending the columns of one to those of the other defines an isometry between their spans: the Gram equality makes the map well-defined and preserves every inner product. An orthogonal extension over the complementary subspaces gives an element of $O(d)$. Thus $W = U_* V$ for some $U_* \in O(d)$.

For any other $Y = UV$, the matrices Y and W are centered and have Gram matrix P/α. Positivity of T gives

$$0 \le \operatorname{tr}\Big(T(W - Y)^\mathsf{T}(W - Y)\Big) = \frac{2}{\alpha}\operatorname{tr} T - 2\operatorname{tr}(TW^\mathsf{T} Y).$$

Because $XP = WT$ and $YP = Y$, the last trace equals $\langle X, Y\rangle_F$. It is at most $\operatorname{tr} T/\alpha$, with equality at $Y = W$. Since $\|X\|_F^2 = \|V\|_F^2 = m$, we obtain

$$d_P(X, V)^2 = 2m - \frac{2}{\alpha}\operatorname{tr} T. \tag{B.15}$$

In particular the infimum is attained in the situation used here.

We now verify uniformly when this situation applies. If $\|X - UV\|_F^2 = e$, then, for each pair of indices,

$$|G_{ij} - (G_\Delta)_{ij}| \le \|x_i - Uv_i\| + \|x_j - Uv_j\| \le 2\sqrt{e}.$$

Taking a sequence approaching the infimum gives

$$\|G - G_\Delta\|_{\mathrm{op}} \le \|G - G_\Delta\|_F \le 2m\, d_P(X, V). \tag{B.16}$$

The bound is independent of d and V.

On the cone of positive semidefinite $m \times m$ matrices, the maps

$$G \longmapsto (\alpha PGP)^{1/2} + (I_m - P), \qquad G \longmapsto \alpha G + J/m$$

are continuous and take the value I_m at G_Δ. Invertibility is an open condition. The second map is positive semidefinite on this cone, so invertibility makes it positive definite. There is therefore an $\varepsilon_m > 0$, in this fixed matrix space, such that both required conclusions hold when $\|G - G_\Delta\|_{\mathrm{op}} < \varepsilon_m$. Together with (B.16), this supplies a positive Procrustes neighborhood depending only on m. We have used continuity of the square root on the positive cone, including at the singular matrix P, but have not differentiated that square root.

B.5 Polynomial square-root coordinates and enclosure

Put $\tau = \rho^2$ and

$$\kappa = \kappa(\tau) = 1 - \frac{\tau}{2m}.$$

If X is on a sufficiently small actual shell $d_P(X, V)^2 = \tau$, (B.15) gives $\operatorname{tr} T = (m-1)\kappa$. Hence

$$D = T - \kappa P \in \mathcal{D}_m. \tag{B.17}$$

Conversely, on the fixed vector space $\mathcal{D}_m$ define the polynomial maps

$$C_\tau(D) = (\kappa P + D)^2 - P, \qquad R_\tau(D) = I_m + \mathcal{U}(C_\tau(D)), \qquad f_\tau(D) = \mathcal{F}_\lambda(R_\tau(D)). \tag{B.18}$$

For (τ, D) near $(0, 0)$, $R_\tau(D)$ is positive definite and $f_\tau(D)$ is jointly C^2. This definition permits all small $D \in \mathcal{D}_m$, irrespective of whether the corresponding Gram matrix is realizable.

Every actual sufficiently small shell is represented by this chart. Indeed, for its D in (B.17),

$$C_\tau(D) = T^2 - P = P(\alpha G - P)P.$$

The matrix $\alpha G - P$ has zero diagonal. The inverse identity (B.11) gives the exact reconstruction

$$R_\tau(D) = \alpha G + J/m. \tag{B.19}$$

There is also a quantitative enclosure that uses no modulus of continuity for the square root. Since the columns of X are unit vectors,

$$\operatorname{tr}(T^2) = \alpha \operatorname{tr}(PGP) = \alpha \left(m - \frac{\left\|\sum_i x_i\right\|^2}{m} \right) \le m - 1,$$
$$\|D\|_F^2 = \operatorname{tr}(T^2) - \kappa^2(m-1) \le (m-1)(1 - \kappa^2) \le \tau.$$

Thus

$$\|D\|_{\mathrm{op}} \le \|D\|_F \le \rho \tag{B.20}$$

for every configuration on every sufficiently small positive shell. The entire feasible shell, not just a chosen smooth path through the simplex, is covered.

B.6 Stationarity and the exact shell Hessian

Conjugation by a permutation matrix preserves P, $\mathcal{D}_m$, the map $\mathcal{U}$, and the chart (B.18). If $X \sim N(0, R)$ and Q is a permutation matrix, then $QX \sim N(0, QRQ^{\mathsf{T}})$, while $\max_i (QX)_i = \max_i X_i$. Together with the equivariance of the chart, this proves that f_τ is permutation invariant. For each sufficiently small fixed τ, its differential at zero is an invariant linear functional on $\mathcal{D}_m$. The average of any $D \in \mathcal{D}_m$ under all permutations is a matrix $aI_m + bJ$. Centeredness gives $a + mb = 0$, while zero trace gives $a + b = 0$. Since $m > 1$, both coefficients vanish. Applying the invariant functional to this average proves

$$\mathrm{d}f_\tau(0)[D] = 0 \qquad (D \in \mathcal{D}_m). \tag{B.21}$$

We compute the Hessian at $(\tau, D) = (0, 0)$. Fix $D \in \mathcal{D}_m$ and write $B = \mathcal{U}(D)$ and $L = \mathcal{U}(D^2)$. The polynomial chart gives exactly

$$R_0(tD) = I_m + 2tB + t^2 L.$$

Here $S(B) = 0$ and, by (B.12), $S(L) = -m \operatorname{tr}(D^2)/2$. Substitution in (B.6) yields

$$f_0(tD) = t^2 Q(D) + O(t^3), \qquad Q(D) = -\frac{mC_\lambda}{2} \operatorname{tr}(D^2) + A_\lambda N(B) - 2(C_\lambda - B_\lambda) R_1(B). \tag{B.22}$$

The $S(A)^2$ term of the full expansion is of order t^4 here. It is needed in the general covariance expansion but contributes nothing to this Hessian. Since f_0 is C^2,

$$\mathrm{d}^2 f_0(0)[D, D] = 2Q(D). \tag{B.23}$$

This follows either by the second derivative chain rule or by comparing the t^2 Taylor coefficient. No remainder symbol is differentiated.

For comparison with the in-plane sectors, write $b = B\mathbf{1}$ and set

$$(B_1)_{ii} = 0, \qquad (B_1)_{ij} = \frac{b_i + b_j}{m - 2} \ (i \neq j), \qquad B_0 = B - B_1.$$

Since $\mathbf{1}^{\mathsf{T}} b = 0$, these matrices satisfy $B_0\mathbf{1} = 0$, $B_1\mathbf{1} = b$, and $\langle B_0, B_1 \rangle_F = 0$. Direct summation also gives $\|B_1\|_F^2 = 2\|b\|^2/(m-2)$. Define

$$E_0 = \frac{\|B_0\|_F^2}{4\alpha}, \qquad E_1 = \frac{\|b\|^2}{m - 2}.$$

They are nonnegative and vanish together only when $D = 0$. Moreover,

$$N(B) = 4\alpha E_0 + 2E_1, \qquad R_1(B) = (m - 2)E_1.$$

For $m = 3$, the symmetric zero-diagonal row-zero matrix B_0 is zero: its three row equations force all three off-diagonal entries to vanish. Thus $E_0 = 0$ in that case.

With

$$\Lambda_0 = \alpha A_\lambda, \qquad \Lambda_1 = \frac{A_\lambda + (m-2)B_\lambda}{2}, \qquad \Lambda_\perp = \frac{m-1}{2} C_\lambda,$$

identities (B.13) and (B.22) give

$$Q(D) = 4[(\Lambda_0 - \Lambda_\perp)E_0 + (\Lambda_1 - \Lambda_\perp)E_1], \qquad \mathrm{d}^2 f_0(0)[D, D] = 8[(\Lambda_0 - \Lambda_\perp)E_0 + (\Lambda_1 - \Lambda_\perp)E_1]. \tag{B.24}$$

The factors in this display are part of the coordinate normalization: the covariance velocity in the present chart is $2B$, and the Hessian is twice the quadratic Taylor coefficient.

The scalar gap assumptions make (B.24) positive definite. One can see the required quantitative coercivity without any additional representation-theoretic argument. Formula (B.13) reduces (B.22) to

$$Q(D) = \left(A_\lambda - \frac{m}{2}C_\lambda\right) N(B) + (2B_\lambda - C_\lambda)R_1(B). \tag{B.25}$$

For $m \geq 4$, Appendix A gives $b_m(\mu) > 1/2$ for $\mu \geq 0$, while the strict normal regime gives $a_m(\mu) > m/2$. Both coefficients in (B.25) are positive, and $N(B) \geq \|D\|_F^2$. For $m = 3$, $R_1(B) = N(B)/2$, so

$$Q(D) = (A_\lambda + B_\lambda - 2C_\lambda)N(B),$$

again with a positive coefficient. In either case there is a $\gamma = \gamma(m, \lambda) > 0$ such that $Q(D) \geq \gamma \|D\|_F^2$ for every $D \in \mathcal{D}_m$.

B.7 Uniform convexity and the equality orbit

Continuity of the Hessian of the joint C^2 map $(\tau, D) \mapsto f_\tau(D)$, together with (B.23), gives positive numbers τ_*, δ, and η, depending only on m and λ, such that

$$\mathrm{d}^2 f_\tau(D)[H, H] \geq \eta \|H\|_F^2 \quad \text{if } |\tau| < \tau_*,\ \|D\|_F < \delta,\ H \in \mathcal{D}_m.$$

Shrink the neighborhoods if necessary to keep the covariance positive definite. The ball in $\mathcal{D}_m$ is convex. Integrating the second derivative along $t \mapsto tD$, and using (B.21), gives

$$f_\tau(D) - f_\tau(0) = \int_0^1 (1-t)\, \mathrm{d}^2 f_\tau(tD)[D, D]\, dt \geq \frac{\eta}{2} \|D\|_F^2 . \tag{B.26}$$

In particular, $f_\tau(D) = f_\tau(0)$ holds if and only if $D = 0$.

Choose $\rho_0 > 0$ smaller than $\sqrt{2m}$, $\sqrt{\tau_*}$, δ, and the uniform Procrustes neighborhood from Subsection B.4. These choices involve only m and λ, before d or V is selected. By (B.20), every actual unit configuration on a shell $0 < \rho < \rho_0$ satisfies the hypotheses of (B.26). By (B.19) and (B.2), its value $f_{\rho^2}(D)$ is its actual decoding deficit.

It remains to verify attainment and identify every equality case. Since $\mathcal{U}(P) = I_m - J$, the point $D = 0$ has

$$R_{\rho^2}(0) = \kappa^2 I_m + (1 - \kappa^2)J, \qquad G_\kappa = \kappa^2 G_\Delta + (1 - \kappa^2)J, \quad \kappa = 1 - \frac{\rho^2}{2m} \in (0, 1).$$

The span of V has dimension $m - 1$, so $d \geq m$ provides a unit $z \perp \operatorname{span}\{v_i\}$. Define

$$N_i = \kappa v_i + \sqrt{1 - \kappa^2}\, z. \tag{B.27}$$

These are unit vectors and have Gram matrix G_κ. Their centered square root is $T = \kappa P$, so K is invertible and (B.15) gives

$$d_P(N, V)^2 = 2m(1 - \kappa) = \rho^2.$$

Thus the minimizer of the extended problem is feasible in the original problem. Its Gram matrix has eigenvalues κ^2/α on H and $m(1 - \kappa^2)$ on $\operatorname{span}\{\mathbf{1}\}$, so it has rank m.

For an arbitrary configuration on this shell, equality in the deficit bound forces $D = 0$ by (B.26), hence $G = G_\kappa$ by reconstruction. The same-Gram isometry argument in Subsection B.4 then gives precisely the $O(d)$ orbit of (B.27). Conversely every configuration on this orbit has the same Gram matrix and success probability. Different choices of the unit normal, and permutations of the regular simplex vertices, produce the same orbit. This proves the equality assertion and the uniform radius statement of Theorem 15.

B.8 The exact value and the sixth-order remainder

Let $\xi \sim N(0, G_\Delta)$ and let b be an independent standard real Gaussian. The vector $\kappa\xi + \sqrt{1-\kappa^2}\, b\mathbf{1}$ has covariance G_κ. For $\kappa \geq 0$, its coordinate maximum is $\kappa \max_i \xi_i + \sqrt{1-\kappa^2}\, b$. Using the Gaussian moment-generating function in (B.1) yields

$$P_{\mathrm{c}}(G_\kappa; \lambda) = \frac{e^{-\lambda^2/2}}{m} e^{\lambda^2(1-\kappa^2)/2} \mathbb{E} e^{\lambda\kappa \max_i \xi_i} = P_{\mathrm{c}}(G_\Delta; \lambda\kappa). \tag{B.28}$$

This proves the exact optimal value, with all prefactors included.

Write $p(s) = P_{\mathrm{c}}(G_\Delta; s)$ using the expectation formula as a real extension in s. This extension is smooth. Indeed, if $M = \max_i \xi_i$, then each coordinate ξ_i is Gaussian and $|M| \leq \max_i |\xi_i|$. Therefore $\mathbb{E}(|M|^k e^{L|M|}) < \infty$ for every fixed k and L. On compact s-intervals these moments dominate all the derivatives of $e^{-s^2/2}\mathbb{E}e^{sM}/m$. In particular p is C^3 near λ, with bounds depending only on m and λ.

Choose $u > 0$ and $M_3 > 0$ such that $|p'''(s)| \leq M_3$ whenever $|s - \lambda| \leq u$. For $h = \lambda\rho^2/(2m)$ small enough, Taylor's theorem gives

$$\left| p(\lambda) - p(\lambda - h) - p'(\lambda)h + \frac{1}{2}p''(\lambda)h^2 \right| \leq \frac{M_3}{6}|h|^3.$$

Together with (B.28), this proves

$$\delta_{\min}(\rho) = \frac{\lambda p'(\lambda)}{2m}\rho^2 - \frac{\lambda^2 p''(\lambda)}{8m^2}\rho^4 + O(\rho^6). \tag{B.29}$$

More explicitly, after decreasing ρ_0 if necessary there is a $C > 0$, depending only on m and λ, such that

$$\left| \delta_{\min}(\rho) - \frac{\lambda p'(\lambda)}{2m}\rho^2 + \frac{\lambda^2 p''(\lambda)}{8m^2}\rho^4 \right| \leq C|\rho|^6, \qquad 0 < \rho < \rho_0. \tag{B.30}$$

For example the Taylor argument permits $C = M_3\lambda^3/(48m^3)$ on a sufficiently small interval. The same estimate holds on a two-sided interval for the real extension $p(\lambda) - p(\lambda(1-\rho^2/(2m)))$.

That exact extension is even in ρ, so every odd derivative at zero vanishes. In particular there is no cubic term; this follows from the exact value, not from dropping an uncomputed cubic remainder. The kernel-checked Taylor argument proves the same quantified $O(\rho^6)$ statement by applying the order-three Taylor little-o theorem, adding back its cubic coefficient to obtain $O(h^3)$ after the quadratic polynomial, and composing with $h = -\lambda\rho^2/(2m)$. It thus verifies the full sixth-order bound, in addition to the parity identity.

The proof uses strict positivity of the existing in-plane curvature gaps. At a threshold where one such gap vanishes, (B.26) is not supplied by this argument. No assertion about that degenerate case, or an effective numerical lower bound for ρ_0, is part of the theorem.

C Lean formalization and reproducibility

C.1 Scope of the certificate

The supplementary Lean development verifies the global quantitative orthant comparison with its explicit constants, the separate centroid penalty, and their consequences for decoding, Procrustes recovery, rank deficiency, unused energy and mean width. This extends the earlier certificate for the ordered channel-parameter crossings, the exact strict-normal minimizer on sufficiently small Procrustes shells, and the quantified $O(\rho^6)$ Taylor remainder. All statements use the actual Gaussian measures, integrals and geometric distance. The global comparison does not assume a witness bound, a product deficit or a desired CDF inequality: these are proved in the dependency chain, including for singular covariances.

The global results hold for $m \geq 2$, with $\lambda > 0$ where a positive decoding coefficient is required. This does not relax the separate hypotheses of the local results: those retain $m \geq 3$, the stated strict curvature gap, and a sufficiently small shell radius. The certificate does not cover numerical diagnostics, learning experiments or historical priority claims. It does not identify finite-radius minimizers in the tangential regimes or on a boundary of equal curvatures. The local radius and remainder constant are proved to exist; no effective numerical value is certified.

The final library entry is `SimplexFull/VerifiedGlobal.lean`. Its dependency closure contains 216 source modules, including 52 vendored foundation modules. A separate audit imports this entry and prints the axioms of 91 selected public declarations. These counts describe the supplied artifact, rather than a claim that every sentence of the manuscript has a formal counterpart. The formal and written proofs may use different intermediate lemmas. The original 94-module local development is retained byte for byte.

C.2 Objects and normalization

The Gaussian density and distribution function are concrete definitions: `SimplexFull.phi` is Mathlib's real standard Gaussian density, and `SimplexFull.Phi` is its Lebesgue integral over $(-\infty, c]$. The definitions of q, h, σ therefore have their usual analytic meaning. For $m \geq 3$, the normalized moments in the crossing theorems are defined using

$$w_{m,\mu}(c) = e^{\mu c}\phi(c)^2\Phi(c)^{m-2}, \qquad Z_m(\mu) = \int_{\mathbb{R}} w_{m,\mu}(c)\,dc,$$

and

$$a_m(\mu) = \frac{\int h(c)^2 w_{m,\mu}(c)\,dc}{Z_m(\mu)}, \qquad b_m(\mu) = \frac{\int \sigma(c) w_{m,\mu}(c)\,dc}{Z_m(\mu)}, \qquad H_m(\mu) = \frac{\int h(c) w_{m,\mu}(c)\,dc}{Z_m(\mu)}.$$

These are respectively `SimplexFull.a`, `SimplexFull.b`, and `SimplexFull.H`; their normalizer is proved positive. The parameter in these integrals is $\mu = \lambda/\sqrt{\alpha}$, where $\alpha = (m-1)/m$. Thus a crossing expressed in μ corresponds to the physical signal amplitude $\lambda = \sqrt{\alpha}\,\mu$. The variables c, μ, and λ are not interchangeable. For $\lambda > 0$,

$$a_m(\mu) = A_\lambda/C_\lambda, \qquad b_m(\mu) = B_\lambda/C_\lambda, \qquad C_\lambda = \frac{\mu e^{-\mu^2/2}}{m} Z_m(\mu) > 0.$$

The value $\mu = 0$ used in the normalized crossing arguments is an endpoint of the moment family, not a division by the zero-amplitude value of C_λ.

The objective `SimplexFull.decodingSuccess` is defined by

$$P_{\mathrm{c}}(G;\lambda) = \frac{e^{-\lambda^2/2}}{m} \int_{\mathbb{R}^m} e^{\lambda \max_i x_i}\, dN(0,G)(x).$$

Here $N(0,G)$ is Mathlib's actual `ProbabilityTheory.multivariateGaussian`, including the singular simplex covariance. Gaussian density identities for positive definite covariances and the common Gaussian shift identity connect the local analytic calculations to this definition. The extended development also identifies this expression with the operational maximum-likelihood decoder for unit signals. For signals of unequal energy it instead uses the class likelihood $\exp(\lambda\langle x_i, z\rangle - \lambda^2\|x_i\|^2/2)$ and the maximum-likelihood/Bayes-value identity. The individual energy correction is retained when proving the unused-energy result.

In the namespace `SimplexFull.Geometry`, `simplexSuccessm` is exactly the function $p_m(\lambda) = P_{\mathrm{c}}(G_\Delta;\lambda)$. The symbols p'_m and p''_m in the Taylor statement are represented by `deriv` and `deriv(deriv(simplexSuccessm))`, rather than by independent coefficient variables.

The theorem `SimplexFull.contDiff_simplexSuccess` supplies genuine C^3 regularity. Its proof establishes all-real exponential integrability of the coordinate maximum and applies Mathlib's moment-generating-function analyticity theorem. In fact, the formal regularity result is stronger than the order needed here.

For configurations, `Geometry.procrustesSqXV` is defined as the real infimum

$$\inf_{U \in O(d)} \sum_{i=1}^{m} \|X_i - UV_i\|^2.$$

The index type consists of linear isometric automorphisms of the whole Euclidean space. Accordingly, `Geometry.orthoOrbitXN` expresses membership in the actual $O(d)$ orbit, including reflections. The distance is not replaced by a chosen alignment or an auxiliary matrix norm. The normal configuration is proved realizable when $d \geq m$, and its squared Procrustes distance is proved to equal ρ^2.

For a finite configuration X, the support of its actual convex hull is defined as the supremum of $\langle g, y \rangle$ over $y \in \operatorname{conv}(\operatorname{range} X)$. The development proves that this supremum is attained and equals the finite maximum of the scores. The normalized sphere measure is the normalization of Mathlib's `volume.toSphere`; it is not defined through a Gaussian ratio. The polar-coordinate measure-preserving map, the radial Gaussian density, and positive homogeneity identify twice its support average (equivalently, the average of $h_K(u)+h_K(-u)$) with the Gaussian expression $2\mathbb{E}\max_i\langle g, X_i\rangle/\mathbb{E}\|g\|$. The denominator is always the Gaussian radial mean in the original ambient dimension d, including when an auxiliary lift uses $d+m$ coordinates.

C.3 Correspondence with the new theorems

Table 4 lists the principal entry declarations. Module paths are relative to `SimplexFull`; all declarations shown below are fully qualified. The package's proof map and source index give the remaining dependency information.

Table 4: Principal mathematical statements and their Lean entry points.

Statement	Module and public declaration
Theorem 2	`GlobalComparison.lean`: `SimplexFull.GlobalComparison.orthant_sqrt_comparison`, `orthant_linear_comparison`, and `quantitative_lowerOrthant` in the same namespace. These use the exact K, κ and all admissible PSD covariances.
Quantitative rectangle and centroid bounds	`GlobalRectangleBound.lean`: `SimplexFull.GlobalRectangle.symmetricRectangle_gap`; `GlobalCentroidConfig.lean`: `SimplexFull.GlobalCentroid.centroid_penalty`.
Theorem 5 and Corollary 6	`GlobalApplicationsCore.lean` and `GlobalApplicationsDecoder.lean`: `SimplexFull.GlobalApplication.decodingSuccess_gram_penalty`, `decodingSuccess_procrustes_penalty`, and `decodingSuccess_rank_deficiency` in that namespace. The centroid and operational unused-energy conclusions are `SimplexFull.GlobalTail.decodingSuccess_centroid_penalty` and `SimplexFull.GlobalTail.ball_decoder_unused_energy`.
Theorem 7	`GlobalSphereGaussianWidth.lean`: `SimplexFull.SphereGaussian.ball_mean_width`. The traditional width-function integral is identified by `GlobalSphereGaussianSymmetry.lean`: `SimplexFull.SphereGaussian.widthIntegral_convexHull_eq_meanWidth`. The Gaussian representation is `SimplexFull.GlobalApplication.ball_mean_width`.

Statement	Module and public declaration
Theorem 13	`GaussianCrossings.lean`: `SimplexFull.gaussian_ordered_crossings`. The strict exclusion of the intermediate coefficient from the minimum is `SimplexFull.gaussian_middle_coefficient_not_min`. The complete chart and its positive-scaling form are in `CrossingSignChart.lean`: `SimplexFull.gaussian_ordered_crossings_sign_chart` and `SimplexFull.gaussian_ordered_crossings_sign_chart_scaled`.
Theorem 14	`GaussianCrossings.lean`: `SimplexFull.gaussian_triangular_unique_crossing`. The equivalent equation $\mu H_3(\mu) = 1$ is established by `SimplexFull.gaussian_triangular_coefficient_eq_iff`.
Theorem 15	`UniformNormalShell.lean`: `SimplexFull.Geometry.gaussian_uniform_exact_normal_shell_m4` and `SimplexFull.Geometry.gaussian_uniform_exact_normal_shell_m3`. The corresponding threshold versions are `SimplexFull.Geometry.gaussian_normal_threshold_uniform_shell_m4` and `SimplexFull.Geometry.gaussian_normal_threshold_uniform_shell_m3`.
Theorem 16	`GaussianTaylor.lean`: `SimplexFull.Geometry.normalTaylor_bound`, `SimplexFull.Geometry.normalTaylor_isBigO`, `SimplexFull.Geometry.gaussian_uniform_taylor_shell_m4`, and `SimplexFull.Geometry.gaussian_uniform_taylor_shell_m3`.

For $m \geq 4$, the crossing declaration asserts $0 < t_1 < t_0 < t_{01}$ and uniqueness on $[0, \infty)$ of the equations

$$a_m(t_1) + (m-2)b_m(t_1) = m - 1, \qquad a_m(t_0) = m/2, \qquad a_m(t_{01}) = m b_m(t_{01}).$$

The declaration `gaussian_ordered_crossings_sign_chart` packages these roots and their uniqueness together with all four open-interval comparisons and all three endpoint comparisons for $L_0, L_1, L_\perp$. Its `gaussian_ordered_crossings_sign_chart_scaled` companion gives the same chart after multiplication by any positive common factor, with the roots independent of that factor. For every $\lambda > 0$ one may therefore use $C_\lambda > 0$ and $\Lambda_j = C_\lambda L_j$ to obtain the physical coefficient chart in Theorem 13. No additional analytic hypothesis is required. The original crossing declaration instantiates the abstract comparison lemmas with the actual Gaussian moments. The proofs include the strict benchmark $b_4(0) > 1/2$, monotonicity, and the integral balance identity $2a_m + (m-1)b_m = m + \mu H_m$. The case $m = 3$ is handled separately: its crossing is $a_3 + b_3 = 2$, and no nonzero $\mathcal{H}_0$ sector is introduced.

The geometric conclusions are packaged as `SimplexFull.Geometry.UniformExactNormalShell` and `SimplexFull.Geometry.UniformTaylorNormalShell`. For fixed m and λ in the strict-normal regime, the latter has the quantifier order

$$\exists C > 0\ \exists \rho_0 > 0\ \forall d \geq m\ \forall V \text{ with Gram } G_\Delta\ \exists z\ \forall\, 0 < \rho < \rho_0\ \forall X \text{ on the feasible shell.}$$

The normal vector z is unit and orthogonal to every V_i. In particular, both C and ρ_0 are independent of the ambient dimension, reference embedding, and competitor. The chosen z is independent of ρ. The final statements contain feasibility of $N_i = \kappa V_i + \sqrt{1-\kappa^2}\, z$, the exact optimal value, a comparison with every feasible X, and equality if and only if `Geometry.orthoOrbitXN`. The strict-normal hypothesis is $a_m(\mu) > m/2$ for $m \geq 4$, and $a_3(\mu) + b_3(\mu) > 2$ for $m = 3$.

The Taylor remainder is the actual function

$$\mathcal{R}_{m,\lambda}(\rho) = p_m(\lambda) - p_m\left(\lambda\left(1 - \frac{\rho^2}{2m}\right)\right) - \left[\frac{\lambda p_m'(\lambda)}{2m}\rho^2 - \frac{\lambda^2 p_m''(\lambda)}{8m^2}\rho^4\right].$$

The declaration `normalTaylor_bound` proves $|\mathcal{R}_{m,\lambda}(\rho)| \le C|\rho|^6$ for $|\rho| < r$, with $C, r > 0$. The companion `normalTaylor_isBigO` states the corresponding two-sided big-O assertion at zero. The generic proof uses a third-order Taylor expansion, obtains an $O(h^3)$ quadratic remainder, and substitutes $h = -\lambda\rho^2/(2m)$. Thus it proves $O(\rho^6)$, rather than only the $o(\rho^4)$ conclusion available from C^2 regularity.

The stronger regularity declaration `SimplexFull.contDiff_simplexSuccess_top` also verifies the analyticity assertion. In the pinned Mathlib version the order type of `ContDiff` is WithTop($\mathbb{N}_\infty$): its outer $\top$ is the analytic order ω, distinct from the smooth order ∞. Applying `ContDiffAt.analyticAt` therefore gives `AnalyticAt` at every real argument. A supplementary kernel-checked statement spells out this consequence for the actual multivariate Gaussian probability integral.

The declaration `SimplexFull.Geometry.normalDeficit_odd_iteratedDeriv_zero` in `TaylorParity.lean` proves that every odd derivative of the exact deficit vanishes at zero. Finally, `SimplexFull.Geometry.simplex_quadratic_coefficient_eq_normal_curvature` in `TaylorCoefficient.lean` identifies the leading coefficient with the actual normal curvature:

$$\frac{\lambda p_m'(\lambda)}{2m} = \frac{m-1}{2}\, C_\lambda.$$

This identity includes the conversion between λ and μ.

C.4 Global proof chain and assumptions

The scalar function Ψ is defined from the actual inverse of $h(s) = s + q(s)$ on the positive axis. Its monotonicity, inverse range, strict concavity, and derivatives are proved. The curvature constant M_0 is the attained maximum of the actual continuous curvature function on the manuscript's compact interval. In particular, the lower Taylor bound used in the centroid argument is not supplied as an external smoothness or remainder assumption.

The quantitative rectangle proof starts from the true bivariate conditional Gaussian law. Differentiation under the compact integral and the boundary derivative give the Plackett formula. Integration on $[0, 1/2]$, monotonicity and evenness yield the stated $\ell^2 e^{-4T^2/3}/(4\pi)$ coefficient. At correlation $+1$ or -1 the residual Gaussian is a Dirac measure and the rectangle mass is the smaller marginal slab mass; no singular density is substituted. The actual Gaussian slab peeling argument and a maximizing off-diagonal pair then give the factor $m(m-1)$ and the exponent $m-2$.

For Theorem 2, existence of actual adaptive witnesses, their compact coordinate bounds, the sum–difference Gaussian law and Fubini's theorem lead to the quantitative doubled-product deficit. The product of the normalized self-convolution factors has Gaussian integral at least one, including for PSD singular covariance by continuous regularization. The fixed-factor rectangle estimate on the central box gives precisely K, with exponent $2m-2$. The correlation-matrix bound $\|R - I\|_F^2 \le m(m-1)$ gives the manuscript's κ by rationalizing the square root. The identity covariance is handled directly. The centroid proof uses the actual PSD variational value, the verified lower Taylor bound for Ψ, and the exact τ, η.

The integration bridge proves

$$\mathbb{E}\max Z - \mathbb{E}\max X = \int_{\mathbb{R}} (F_X(c) - F_Z(c))\, dc$$

from a centered signed-tail kernel and Fubini, with integrability proved. Applied to the exponential of the maximum it retains the exact window factor $e^\mu - 1$. Gaussian normalization at $R = \alpha G + J/m$ supplies the factors α^2 and $\sqrt{\alpha}$. Consequently the final entries use

$$c_g = W\kappa\alpha^2, \quad c_s = W\eta, \quad c_P = \left(\frac{\alpha}{c_g} + \frac{1}{mc_s}\right)^{-1}, \qquad b_g = \kappa\alpha^{3/2}, \quad b_s = \eta/\sqrt{\alpha}.$$

None of those entries retains a unit-comparison or target-CDF hypothesis.

The geometric inequality is proved for vectors of arbitrary lengths and for the actual infimum over $O(d)$; a polar-alignment witness also proves attainment. The rank corollary assumes the actual matrix rank bound, with its bound independent of any realization dimension. For ball configurations the formal proof uses mutually orthogonal extra coordinates. Adding the resulting independent centered Gaussian noise cannot decrease the expected maximum; the corresponding likelihood argument proves the operational decoding comparison for nonunit signals. This orthogonal lift is an alternative to the shared-coordinate lift in the written unused-energy proof.

The foundation sources from the weak-simplex and simplex-uniqueness developments are supplied with exact upstream commits, per-file provenance and their original licenses and attributions in `VENDOR_PROVENANCE.json` and `licenses/`. Their imported closure is included in the same fresh trust-level-zero build as the new modules, rather than accepted through inherited project binaries.

C.5 Build and axiom audit

The released archive is `concrete-simplex-lean-global-20261106.zip`. Its SHA-256 digest is

`4e8e24831e8c56ab07523aed22c89c4ba6854726accccff2aa32c0587133896b`.

The toolchain is Lean 4.31.0, pinned by `leanprover/lean4:v4.31.0`; the Mathlib revision is

`fabf563a7c95a166b8d7b6efca11c8b4dc9d911f`.

The successful recorded run is `20261106-180522-470580`. Its `evidence/verification.json` records all 216 source-module commands and exit codes, source and compiled-file hashes, the successful combined import, and the final axiom audit. All source hashes remained unchanged through verification. The manuscript bundle also supplies the independent Grok Build and agent cross-reviews.

The verifier copies the three project-source namespaces into a fresh snapshot and creates an initially empty output directory. It compiles their explicit dependency closure in dependency order. All inherited project build paths, including paths containing only a bare root module, are excluded from `LEAN_PATH`; the pinned Mathlib cache is supplied by Lake. Every project source is compiled with Lean's trust level zero:

```
lean -j 2 -t 0 -R <snapshot/src> \
  -o <fresh/build/Module.olean> \
  <snapshot/src/Module.lean>
```

The final audit imports `SimplexFull.VerifiedGlobal` and invokes `#printaxioms` for 91 selected declarations. Their dependency reports contain only `propext`, `Classical.choice`, and `Quot.sound`. Acceptance requires exactly one recognized report per requested declaration. Missing, duplicate, unexpected or nonstandard reports cause verification to fail.

After removing comments and strings, source scanning rejects `sorry`, `admit`, project `axiom` declarations, `unsafe`, `native_decide` and `debug.skipKernelTC`. Import parsing rejects unknown project roots and unsupported header layouts instead of silently omitting them. Positive and negative regression tests cover the parser, binary-path isolation and axiom reports. These checks accompany, rather than replace, kernel checking.

C.6 Reproducing the verification

After extracting the archive, install the pinned Lean toolchain with `elan` and make Python available. From the package directory run

```
lake update
lake exe cache get
lake env python reproduce.py
```

The Mathlib revision is fixed in `lakefile.toml`. The verifier uses `SimplexFull.Verified Global` and `AUDIT_THEOREMS_GLOBAL.json`, creates a fresh timestamped build, and returns a nonzero status on any source, compilation or axiom-audit failure. The package supplies `PROOF_MAP_GLOBAL.md`, `SOURCE_INDEX.json`, `SHA256SUMS.json`, the individual compiler logs, upstream provenance and independent cross-reviews. The source, compiler output and `evidence/verification.json` are the reproducibility evidence.

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